
UNIFYING FORMAL EXPLANATIONS: A COMPLEXITY-THEORETIC PERSPECTIVE

Anonymous authors
Anonymous submission

ABSTRACT

This paper makes the contribution structure of formal explanation computation explicit. We study two central explanation families for ML predictions: (i) *sufficient reasons*, feature subsets that determine a prediction when fixed, and (ii) *contrastive reasons*, feature subsets whose modification can alter a prediction. Our first contribution is a unified framework showing that non-probabilistic and probabilistic, local and global variants of these explanations can be expressed as minimization problems over a common probabilistic value function. Our second contribution is to identify the combinatorial properties that govern the resulting algorithms: (i) *monotonicity*, (ii) *submodularity*, and (iii) *supermodularity*. Our third contribution is a structural separation between local and global explanations: the *local* value functions lack these properties, whereas the corresponding *global* value functions satisfy them. This separation yields our main complexity contribution: polynomial-time and approximation results for global explanations with provable guarantees across neural networks, decision trees, and tree ensembles, contrasted with NP-hardness for even simplified local counterparts.

1 INTRODUCTION

Formal explanations are most useful when they come with guarantees, but the first obstacle for such guarantees is computational. Prior work repeatedly shows that many desirable explanation types are hard to compute (Barceló et al., 2020; Van den Broeck et al., 2022), and this difficulty typically becomes sharper for complex or highly non-linear models (Barceló et al., 2020; Adolfi et al., 2025; 2024). This paper contributes to that complexity-theoretic program by asking which explanation guarantees become tractable once the problem is framed at the right level of generality. The broader literature has mapped several boundaries between efficiently computable and intractable explanations across model classes (Barceló et al., 2020; Wäldchen et al., 2021; Arenas et al., 2022; 2023; Marzouk & De La Higuera, 2024; Ordyniak et al., 2023; Laber, 2024; Bhattacharjee & Luxburg, 2024; Blanc et al., 2021; 2022); our contribution is to expose a common value-function structure behind several of these boundaries.

From sufficient to contrastive reasons. We focus on two explanation families that anchor much of the formal XAI complexity literature: (i) *sufficient reasons* and (ii) *contrastive reasons* (Barceló et al., 2020; Arenas et al., 2022; Audemard et al., 2022a; Arenas et al., 2021; Barceló et al., 2025; Marques-Silva & Ignatiev, 2022; Ignatiev et al., 2020b; Darwiche & Hirth, 2020). A *sufficient reason* is a subset of input features S such that fixing these features fixes the model prediction, regardless of how the complementary set $\bar{S}$ is assigned. A *contrastive reason* is a subset of input features S such that modifying these features changes the model prediction.

Unlike additive attribution methods, which allocate importance scores across features but are often hard for humans to interpret (Kumar et al., 2020) or lack actionability (Bilodeau et al., 2024), sufficient and contrastive reasons provide *discrete, condition-like explanations* that directly answer “what is enough to justify this prediction?” or “what must change to flip it?”. This set-based form is why they serve as the paper’s central objects: they have a central role in classic XAI methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dhurandhar et al., 2018), can improve human prediction over additive models (Ribeiro et al., 2018; Yin & Neubig, 2022; Dhurandhar et al., 2018), and support downstream tasks such as bias detection (Balkir et al., 2022; Carter et al., 2021; La Malfa et al., 2021; Muthukumar et al., 2018), model debugging (Jacovi et al., 2021), and anomaly detection (Davidson

et al., 2025). This directly motivates the minimality focus of our contribution: smaller sufficient and contrastive explanations are generally considered more interpretable, so *minimality* becomes the key guarantee whose computational status we study (Ribeiro et al., 2018; Lopardo et al., 2023; Carter et al., 2019; Barceló et al., 2020; Arenas et al., 2022; Blanc et al., 2021; Wäldchen et al., 2021).

Barceló et al. (2020) conducted one of the earliest studies on the complexity of deriving sufficient and contrastive reasons. Their work established that finding the minimal-sized sufficient reason for a decision tree is NP-hard, with the complexity further increasing to Σ_2^P -hard for neural networks. In the case of contrastive reasons, they demonstrated that computing the smallest possible explanation is solvable in polynomial time for decision trees but becomes NP-Hard for neural networks. Similar hardness results were later shown to hold for tree ensembles as well (Izza & Marques-Silva, 2021; Ordyniak et al., 2024; Audemard et al., 2022b).

From non-probabilistic to probabilistic explanations. A common criticism of the classic definition of sufficient and contrastive reasons is their rigidity and lack of flexibility, as they hold in an absolute sense over entire domains and can thus lead to excessively large or uninformative explanations (Ignatiev et al., 2019; 2020a; Arenas et al., 2022; Wäldchen et al., 2021). To address these limitations, the literature has shifted towards a more general definition that incorporates a *probabilistic* perspective on sufficient and contrastive reasons (Wäldchen et al., 2021; Arenas et al., 2022; Blanc et al., 2021; Izza et al., 2023; Xue et al., 2023). Under this framework, the goal is to identify subsets of input features that influence a prediction with a probability exceeding a given threshold δ .

Wäldchen et al. (2021) were the first to study the complexity of probabilistic sufficient reasons, showing that for CNF classifiers the problem is NP^{PP} -hard. This hardness extends to tree ensembles and neural networks (Barceló et al., 2020; Ordyniak et al., 2023). A central theoretical insight in the probabilistic setting is the *lack of monotonicity* in the probability function (Arenas et al., 2022; Izza et al., 2023), which makes even *subset minimal* explanations computationally hard. Strikingly, (Arenas et al., 2022) show that finding a subset minimal probabilistic sufficient reason is NP-hard even for decision trees — unlike in the non-probabilistic case, where such explanations are computable in polynomial time (Huang et al., 2021; Barceló et al., 2020).

From local to global explanations. The third axis is the distinction between local and global explanation goals. Recent work by Bassan et al. (2024) moves from the *local* non-probabilistic setting, where an explanation targets one prediction, to the *global* non-probabilistic setting, where the explanation must characterize behavior over an entire domain. That shift is central to our paper, but the non-probabilistic global criterion remains extremely strict: as in other absolute-guarantee approaches (Ignatiev et al., 2019; Barceló et al., 2020; Arenas et al., 2021; Darwiche & Hirth, 2020), excluding a feature from a global sufficient reason means the feature is strictly redundant (Bassan et al., 2024). In practice, this can make the explanation include nearly all input features.

OUR CONTRIBUTIONS

1. **Unified task formulation.** We show that sufficient and contrastive explanations, in probabilistic and non-probabilistic as well as local and global forms, can be written as one minimization problem over a unified value function. This formulation makes three complexity-relevant properties explicit: (a) *monotonicity*, (b) *supermodularity*, and (c) *submodularity*.
2. **Structural separation.** We prove that the same properties behave differently in local and global explanations. The *local* value functions for sufficient and contrastive reasons are *non-monotonic*, whereas their *global* counterparts are *monotonic non-decreasing*. Under feature independence, the global sufficient value function is *supermodular* and the global contrastive value function is *submodular*, while the corresponding local properties fail.
3. **Subset-minimal complexity results.** We use the structural separation to derive tractability results for global subset-minimal explanations and hardness results for local ones. Across (i) neural networks, (ii) decision trees, and (iii) tree ensembles, we show that global subset-minimal sufficient/contrastive explanations can be computed in polynomial time in settings where the corresponding local probabilistic explanations remain NP-hard, including the decision-tree setting of (Arenas et al., 2022). We also extend the positive result to black-box models under empirical distributions.
4. **Cardinal-minimal approximation frontier.** Finally, we exploit global submodularity and supermodularity to obtain provable *constant-factor* approximation guarantees for cardinally

minimal global explanations when the empirical distribution is fixed, including for neural networks and tree ensembles. We contrast this with local inapproximability: no *bounded* approximation is possible even under very simple assumptions.

The main text is organized to foreground these contributions: it states the unifying framework, the value-function properties, and the resulting complexity consequences, while the appendix provides the full proofs.

2 PRELIMINARIES

Setting. We first fix the notation used by the unified formulation. The input space has dimension $n \in \mathbb{N}$, with vectors $\mathbf{x} := (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. Each coordinate $\mathbf{x}_i$ ranges over a domain $\mathcal{X}_i$, which may be discrete or continuous, so the full feature space is $\mathbb{F} := \mathcal{X}_1 \times \mathcal{X}_2 \times \dots \times \mathcal{X}_n$. We study classifiers $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ is the number of classes, together with a *generic* distribution $\mathcal{D} : \mathbb{F} \rightarrow [0, 1]$. Because the true distribution may be computationally unavailable, several of our complexity results also use a fixed empirical dataset $\mathbf{D} := \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$ as a tractable proxy, for example when $\mathbf{D}$ is sampled from the training data.

The local/global distinction is the main structural axis of the paper. A local explanation targets a specific input $\mathbf{x} \in \mathbb{F}$ and explains why the model outputs $f(\mathbf{x})$ on that instance. A global explanation is input-independent and is intended to summarize how f behaves across a broader region, or all, of the input space.

Models. Our value-function properties are stated generally, but our complexity contribution is instantiated on model classes that span the interpretability spectrum. We analyze (i) decision trees, (ii) neural networks (all architectures at least as expressive as feed-forward ReLU networks), and (iii) tree ensembles, including majority-voting ensembles (e.g., random forests) and weighted-voting ensembles (e.g., XGBoost). Formal definitions appear in the Appendix.

Distributions. Distributional assumptions are part of the contribution statement because they determine which value-function properties and algorithms apply. We consider three cases: (i) *general distributions*, which impose no structural assumptions and are used for universal properties; (ii) *empirical distributions*, derived from a finite dataset $\mathbf{D}$ and commonly used in XAI (Lundberg & Lee, 2017; Van den Broeck et al., 2022); and (iii) *independent distributions*, where features are mutually independent, a standard assumption in several XAI analyses (Arenas et al., 2022; 2023; Lundberg & Lee, 2017; Ribeiro et al., 2018). Empirical distributions need not be independent; they may encode dependencies present in the finite dataset (Van den Broeck et al., 2022). Complete formal definitions are deferred to the Appendix.

3 FORMS OF REASONS

This section makes explicit how the paper’s explanation families fit into one framework. We begin with strict non-probabilistic local and global sufficient/contrastive reasons, then relax them to probabilistic variants that are later captured by the unified value functions.

3.1 Non-Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

Sufficient reasons. In the context of *feature selection*, users often select the top k features that contribute to a model’s decision. We examine the well-established *sufficiency* criterion for this selection, which aligns with commonly used explainability methods (Ribeiro et al., 2018; Carter et al., 2019; Ignatiev et al., 2019; Dasgupta et al., 2022). This feature selection can be carried out either locally — focusing on a single prediction — or globally — across the entire input domain. Following standard conventions, we define a *local sufficient reason* as a subset of features $S \subseteq \{1, \dots, n\}$ such that when the features in S are fixed to their corresponding values in $\mathbf{x}$, the model’s prediction remains $f(\mathbf{x})$ regardless of the values assigned to the remaining features $\bar{S}$. Formally, S is a local sufficient reason for $\langle f, \mathbf{x} \rangle$ iff the following condition holds:

$$\forall \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (1)$$

Here, $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ denotes the hybrid vector that keeps the features in S from $\mathbf{x}$ and takes the remaining features from $\mathbf{z}$. This definition is the local sufficiency building block used throughout the paper. It is also closely related to *semi-factual* explanations (Kenny & Keane, 2021; Alfano et al., 2025), which search for alternative inputs $\mathbf{z}'$ that preserve the prediction (i.e., $f(\mathbf{z}') = f(\mathbf{x})$) while remaining close to the original point; with $\mathbf{z}' = (\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ and Hamming distance, the notions coincide.

The global counterpart strengthens the local definition in a way that later drives our structural results. A *global sufficient reason* (Bassan et al., 2024) is a subset $S \subseteq \{1, \dots, n\}$ that is a local sufficient reason for *every* input $\mathbf{x} \in \mathbb{F}$:

$$\forall \mathbf{x}, \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}). \quad (2)$$

Contrastive reasons. The contrastive side of the framework asks which feature subsets can *modify* a prediction (Dhurandhar et al., 2018; Mothilal et al., 2020; Guidotti, 2024). Its minimality goal is to identify small sets of coordinates whose change can alter the output. Formally, $S \subseteq \{1, \dots, n\}$ is a *local contrastive reason* for $\langle f, \mathbf{x} \rangle$ iff:

$$\exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (3)$$

This local contrastive definition is the formal analogue of a counterfactual explanation (Guidotti, 2024; Mothilal et al., 2020): it seeks a nearby assignment $\mathbf{x}'$ for which the prediction flips, i.e., $f(\mathbf{x}') \neq f(\mathbf{x})$. With Hamming weight as the distance metric, the two notions coincide.

As with sufficiency, the global contrastive version requires the subset to support prediction changes across the domain of interest. This definition connects both to bias-identification approaches (Arenas et al., 2021; Bassan et al., 2024; Darwiche & Hirth, 2020) and to *group* counterfactual methods that search for counterfactuals over multiple instances (Carrizosa et al., 2024; Warren et al., 2024). Formally, S is a *global contrastive reason* for f iff:

$$\forall \mathbf{x} \in \mathbb{F}, \quad \exists \mathbf{z} \in \mathbb{F}, \quad f(\mathbf{z}_S; \mathbf{x}_{\bar{S}}) \neq f(\mathbf{x}). \quad (4)$$

3.2 Probabilistic SUFFICIENT AND CONTRASTIVE REASONS

The probabilistic definitions are the bridge from rigid formal guarantees to the unified optimization task. Because non-probabilistic sufficient and contrastive reasons can impose overly strict conditions, prior work relaxes them by requiring the relevant guarantee to hold with probability at least $\delta \in [0, 1]$ (Arenas et al., 2022; Ribeiro et al., 2018; Izza et al., 2023; Wäldchen et al., 2021; Bounia & Koriche, 2023; Subercaseaux et al., 2025; Wang et al., 2021a; Blanc et al., 2021). The special case $\delta = 1$ recovers the original non-probabilistic definitions.

Probabilistic sufficient reasons. The local probabilistic sufficient value fixes the features in S to their values in $\mathbf{x}$ and samples the remaining features from $\mathcal{D}$. A subset $S \subseteq \{1, \dots, n\}$ is a *local δ -sufficient reason* for $\langle f, \mathbf{x} \rangle$ when the prediction remains unchanged with probability at least δ :

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \geq \delta. \quad (5)$$

where $\mathbf{z}_S = \mathbf{x}_S$ means that the coordinates in S of $\mathbf{z}$ are fixed to match $\mathbf{x}$. For the global version, we follow the standard averaging view of global explanations and define $S \subseteq \{1, \dots, n\}$ as a *global δ -sufficient reason* for $\langle f \rangle$ when the expected local sufficiency probability over $\mathcal{D}$ is at least δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \geq \delta. \quad (6)$$

Probabilistic contrastive reasons. The contrastive probability is defined dually. We fix the complementary set $\bar{S}$ to its values in $\mathbf{x}$, allow the features in S to vary according to $\mathcal{D}$, and require a prediction change with probability at least δ . Formally:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}) \geq \delta. \quad (7)$$

The global contrastive definition is parallel and completes the four-way explanation family used in the paper. A subset S is a *global δ -contrastive reason* when the expectation of the local contrastive value under $\mathcal{D}$ is at least δ :

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (8)$$

4 FORMS OF MINIMALITY

Minimality is the guarantee that turns the definitions above into computational problems. Across sufficient and contrastive, local and global explanations, smaller subsets are typically considered more meaningful; we therefore analyze two minimality notions for every explanation type in the framework:

Definition 1. Given a subset $S \subseteq \{1, \dots, n\}$ that already satisfies the relevant explanation condition, we distinguish:

1. S is a *cardinally-minimal explanation* (Barceló et al., 2020; Bassan et al., 2024) iff S has the smallest explanation cardinality $|S|$ (i.e., there is no explanation S' such that $|S'| < |S|$).
2. $S \subseteq \{1, \dots, n\}$ is a *subset-minimal explanation* (Arenas et al., 2022; Ignatiev et al., 2019) iff S is an explanation, and any $S' \subsetneq S$ is not an explanation.

Cardinal minimality is strictly stronger than subset minimality: every cardinally minimal S is subset-minimal, but not conversely (Appendix B.3 gives an example). We use the terms subset and cardinal minimality, rather than local and global minima, to avoid confusion with local versus global explanations. Both minimality notions apply to *all* explanation forms in the paper. For example, a cardinally minimal local probabilistic δ sufficient reason has minimum possible $|S|$, whereas a subset-minimal one only requires that no proper subset $S' \subsetneq S$ also qualifies.

5 A UNIFIED COMBINATORIAL OPTIMIZATION TASK

We now state the unifying contribution formally. Every explanation problem introduced above — local or global, sufficient or contrastive, probabilistic or non-probabilistic — can be written as finding a subset S with $v(S) \geq \delta$, with $\delta := 1$ in the non-probabilistic case. Let v_{suff}^ℓ be the local sufficiency probability from equation 5, i.e., $\Pr_{\mathbf{z} \sim \mathcal{D}_p}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, and let v_{suff}^g be its global variant from equation 6. Likewise, v_{con}^ℓ and v_{con}^g denote the local and global contrastive probabilities from equations 7 and 73. With this notation, cardinal-minimal explanation computation becomes:

Cardinally Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and $|S|$ is minimal.

The subset-minimal version uses the same value-function interface but weakens the output requirement from globally smallest size to deletion-wise minimality:

Subset Minimal δ -Explanation:

Input: Model f , a distribution $\mathcal{D}$, (possibly, an input $\mathbf{x}$), a *general* value function $v : 2^n \rightarrow [0, 1]$ (defined using f , $\mathcal{D}$, and possibly $\mathbf{x}$), and some $\delta \in [0, 1]$.

Output: A subset $S \subseteq [n]$ such that $v(S) \geq \delta$ and for any $S' \subsetneq S$ it holds that $v(S') < \delta$.

Properties of v that affect the complexity. The next contribution is to identify which properties of v make the unified task tractable or hard. The first property is *non-decreasing monotonicity*: adding a feature never decreases the value, so the marginal contribution $v(S \cup \{i\}) - v(S)$ is always non-negative. Formally:

Definition 2. We say that a value function v maintains non-decreasing monotonicity if for any $S \subseteq \{1, \dots, n\}$ and any $i \in \{1, \dots, n\}$ it holds that: $v(S \cup \{i\}) \geq v(S)$.

The second and third properties, *supermodularity* and *submodularity*, describe how marginal gains change as the selected set grows. Supermodularity means the marginal contribution $v(S \cup \{i\}) - v(S)$ is monotone non-decreasing in S ; submodularity is the dual diminishing-returns condition. These properties are the reason global sufficient and contrastive explanations admit different algorithmic guarantees. Formally:

Definition 3. Let there be some value function v , some $S \subseteq S' \subseteq \{1, \dots, n\}$, and $i \notin S'$. Then:

1. v maintains supermodularity iff it holds that: $v(S \cup \{i\}) - v(S) \leq v(S' \cup \{i\}) - v(S')$.
2. v maintains submodularity iff it holds that: $v(S \cup \{i\}) - v(S) \geq v(S' \cup \{i\}) - v(S')$.

6 UNRAVELING NEW PROPERTIES OF THE GLOBAL VALUE FUNCTIONS

We now state the structural contribution that drives the complexity results. In local non-probabilistic explanation computation, subset-minimal sufficient or contrastive reasons are computable because monotonicity holds in the special case $\delta = 1$. That case is restrictive, and prior work shows that once one moves to probabilistic local guarantees for arbitrary δ , monotonicity breaks down and the corresponding problems become harder (Arenas et al., 2022; Kozachinskiy, 2023; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b).

The global non-probabilistic setting has an even stronger but highly rigid property: Bassan et al. (2024) show that there is exactly *one* subset-minimal explanation. Because this result requires $\delta = 1$, the global condition must hold for *all* inputs, and the unique subset coincides with all features that are not strictly redundant (Bassan et al., 2024). Our first new structural result shows that this uniqueness does not survive the probabilistic global setting: for arbitrary δ , the number of subset-minimal global sufficient reasons can be exponential.

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

The failure of uniqueness does not remove the algorithmic structure we need. Our second structural result is that global value functions remain monotone for every δ , both for sufficiency (v_{suff}^g) and contrastiveness (v_{con}^g). This is the key local/global separation: the corresponding local functions (v_{suff}^ℓ and v_{con}^ℓ) do not satisfy monotonicity.

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Monotonicity gives subset-minimal tractability; the stronger approximation results come from curvature-like structure. Under feature independence, we prove that the global sufficient value function v_{suff}^g is supermodular. The local sufficient value function v_{suff}^ℓ lacks this property even under the more restrictive uniform distribution. Specifically:

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.*

For global probabilistic *contrastive* reasons, the complementary structural contribution is submodularity rather than supermodularity. This again separates global from local explanations: the local contrastive value function is neither submodular nor supermodular.

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^ℓ is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

7 COMPUTATIONAL COMPLEXITY RESULTS

7.1 SUBSET MINIMAL EXPLANATIONS

We now turn the structural properties into complexity results. For *subset minimal* explanations, the decisive property is monotonicity: the previous section shows that v_{con}^g and v_{suff}^g are monotone, whereas v_{con}^ℓ and v_{suff}^ℓ are not. This difference explains why the same greedy deletion principle yields subset-minimal global explanations but cannot guarantee subset minimality locally. The resulting contribution is a tractability/hardness separation: several global subset-minimal explanations

are polynomial-time computable, while their local counterparts remain hard. We begin with the generalized greedy algorithm:

Algorithm 1 Subset Minimal Explanation Search

Input Value function v , and some $\delta \in [0, 1]$

```

1:  $S \leftarrow \{1, \dots, n\}$ 
2: while  $\min_{i \in S} v(S \setminus \{i\}) \geq \delta$  do
3:    $j \leftarrow \operatorname{argmax}_{i \in S} v(S \setminus \{i\})$ 
4:    $S \leftarrow S \setminus \{j\}$ 
5: end while
6: return  $S$ 

```

$\triangleright S$ is a (subset minimal?) δ -explanation

Algorithm 1 implements the subset-minimal search used in our positive global results. It starts from the full feature set $S = \{1, \dots, n\}$ and repeatedly removes a feature whenever the remaining set still has value at least δ . Among removable candidates, it selects the feature j that maximizes $v(S \setminus \{j\})$, i.e., the removal that preserves the largest value. The final set is returned once no single deletion preserves the threshold.

The correctness contribution is exactly where monotonicity enters. At termination, every one-feature deletion violates the threshold. If v is monotone, then deleting any larger subset can only reduce the value further, so the returned set is genuinely subset-minimal. Applying the monotonicity results above yields:

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

This proposition is the bridge from the unified framework to model-specific complexity. We next instantiate it for the model classes and distributional assumptions introduced above.

Decision trees. We begin with an interpretable model class to make the separation especially clear. Let f be a decision tree and let $\mathcal{D}$ be independent. In this setting, Arenas et al. (2022) show that, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm computes a subset-minimal local δ -sufficient reason for decision trees under independent distributions, even under the uniform distribution. Our contribution is the global counterpart: under the stated computability condition for the probability term, a subset-minimal global δ -sufficient reason for decision trees is computable in polynomial time. Formally:

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Extension to other tractable models. The same global tractability logic also extends to *orthogonal DNFs* (Crama & Hammer, 2011), i.e., DNF formulas whose terms are pairwise mutually exclusive. This extension is useful because orthogonal DNFs *generalize* decision trees while retaining a structure that supports efficient value computation. Appendix N gives the proof.

Neural networks, tree ensembles, and other complex models. We next show that the positive global result is not confined to interpretable models. Under empirical distributions, and assuming the model f supports polynomial-time inference, subset-minimal global δ -sufficient and δ -contrastive reasons can be computed in polynomial time.

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Because this positive result holds for *any* model under an empirical distribution, it yields a sharper separation from the local setting. For neural networks and tree ensembles, subset-minimal *local* explanations remain intractable even under empirical distributions, while the global versions are polynomial-time computable:

Theorem 2. Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $\text{PTIME}=\text{NP}$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.

7.2 APPROXIMATE CARDINALLY MINIMAL EXPLANATIONS

We now move to the stronger guarantee of cardinal minimality. Here the goal is not merely deletion-wise minimality, but the smallest possible explanation size. The contribution of this subsection is to show that the global submodularity and supermodularity proved earlier are strong enough to yield approximation guarantees, while local explanations admit no bounded approximation in the corresponding settings.

A unified greedy approximation algorithm. The approximation results again use the unified value-function view. For monotone submodular functions, cardinal-minimal explanation search aligns with *submodular set cover* (Wolsey, 1982); for monotone supermodular functions, it aligns with a non-submodular variant (Shi et al., 2021). Both have known approximation analyses (Wolsey, 1982; Iyer & Bilmes, 2013; Chen & Crawford, 2023). Algorithm 2 is the greedy procedure to which we apply those analyses for contrastive and sufficient reasons, respectively.

Algorithm 2 Cardinally Minimal Explanation Approximation Search

Input Value function v , and some $\delta \in [0, 1]$

- 1: $S \leftarrow \emptyset$
- 2: **while** $\max_{i \in S} v(S \cup \{i\}) < \delta$ **do**
- 3: $j \leftarrow \operatorname{argmax}_{i \in S} v(S \cup \{i\})$
- 4: $S \leftarrow S \cup \{j\}$
- 5: **end while**
- 6: **return** S $\triangleright |S|$ is a (provable approximation?) of a cardinally minimal δ -explanation

Algorithm 2 mirrors Algorithm 1 but proceeds bottom-up. It starts with the empty set and incrementally adds the feature that gives the largest value increase, stopping once the threshold δ is reached. Its guarantees depend on whether the relevant global value function is submodular or supermodular.

Cardinally minimal contrastive reasons. For global contrastive reasons, monotonicity and submodularity reduce the task to classic *submodular set cover*, so we can apply Wolsey (1982)’s guarantee to Algorithm 2. For integer-valued objectives this gives a Harmonic-based factor, and more generally an $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation. Under an *empirical* distribution $\mathcal{D}$ with fixed sample size, the factor is constant and logarithmic in the sample size. The local setting has the opposite behavior: even with a *single* sample point, no *bounded* approximation exists.

Theorem 3. Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.

To clarify the tightness of this contribution, we also provide a matching *lower bound* for Theorem 3 in the case of an empirical distribution without an independence assumption (Appendix M).

Cardinally minimal sufficient reasons. For sufficient reasons, the relevant global property is supermodularity, so the approximation argument uses the supermodular set-cover variant rather than the submodular one. This problem is harder, but it has guarantees under bounded curvature (Shi et al., 2021). The *total curvature* of $v : 2^n \rightarrow \mathbb{R}$ is $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus i)}{v(i) - v(\emptyset)}$. Using Shi et al. (2021), we show that Algorithm 2 achieves an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v([n])}{\min_{i \in [n]} v(i)}\right)\right)$ -approximation.

Under a fixed empirical distribution, this approximation becomes constant. The sufficient-reason bound is weaker than the contrastive $\mathcal{O}(\ln(|D|))$ bound because it includes the additional $\frac{1}{1-k^f}$ factor,

but it still gives a constant-factor global approximation. The local sufficient variant again remains inapproximable, with no *bounded* approximation even when $|D| = 1$.

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\frac{1}{1-kf} + \ln\left(\frac{v_{\text{diff}}^g([n])}{\min_{i \in [n]} v_{\text{diff}}^g(\{i\})}\right)\right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME}=\text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Overall, these findings strengthen Subsection 7.1, which showed the tractability of computing subset-minimal global explanations in stark contrast to local ones. Here, we further show that approximating cardinally minimal global explanations is tractable, unlike their inapproximable local counterparts.

8 LIMITATIONS AND FUTURE WORK

The contribution of this paper is structural and complexity-theoretic, but it is not exhaustive. Some results hold for general value functions and distributions, while others are instantiated for particular model classes, distributional assumptions, or explanation definitions. This leaves several reviewer-relevant boundaries for future work.

More specifically, the framework foregrounds three open theoretical directions:

1. First, the global tractability results may be sharpened for simpler model classes such as decision trees, linear models, or other models with inherently tractable structure. Although local explanations for such models are often intractable (Arenas et al., 2022; Subercaseaux et al., 2025), the global explanation forms identified here have additional structure that may support tighter complexity and approximation bounds, especially under empirical distributions.
2. Second, when *exact* expectations are infeasible, the framework raises the question of whether approximation tools — such as Fully Polynomial Randomized Approximation Schemes for DNF formulas (Meel et al., 2019) and thus tree ensembles, or Monte Carlo approximations of the kind used in (Subercaseaux et al., 2025) — can preserve the same minimality guarantees. The open issue is how much approximation error can be tolerated without invalidating the explanation guarantee.
3. Third, the same local-to-global structural pattern may appear for other feature-importance objectives, such as losses based on KL divergence (Conmy et al., 2023). If analogous monotonicity, submodularity, or supermodularity properties hold, the contribution could extend to efficient feature selection under alternative metrics.

9 CONCLUSION

This paper contributes a unified complexity-theoretic account of sufficient and contrastive explanations. First, we formulate local/global and probabilistic/non-probabilistic variants as minimization problems over common value functions. Second, we prove a structural local/global separation: local value functions lack the monotonicity, submodularity, and supermodularity needed by standard combinatorial algorithms, whereas global value functions exhibit monotonicity, global contrastive values are submodular, and global sufficient values are supermodular. Third, we turn these properties into complexity consequences: global explanations with provable guarantees can be computed or approximated efficiently across model classes including neural networks, decision trees, and tree ensembles, while corresponding local explanations remain NP-hard or inapproximable even in simplified settings. Together, these results make the tractable and intractable frontiers of formal explanations more explicit.

REFERENCES

Federico Adolphi, Martina G Vilas, and Todd Wareham. Complexity-Theoretic Limits on the Promises of Artificial Neural Network Reverse-Engineering. In *Proc. 46th Annual Meeting of the Cognitive Science Society (CogSci)*, 2024.

486 Federico Adolphi, Martina Vilas, and Todd Wareham. The Computational Complexity of Circuit
487 Discovery for Inner Interpretability. In *Proc. 13th Int. Conf. on Learning Representations (ICLR)*,
488 2025.

489

490 Gianvincenzo Alfano, Sergio Greco, Domenico Mandaglio, Francesco Parisi, Reza Shahbazian, and
491 Irina Trubitsyna. Even-If Explanations: Formal Foundations, Priorities and Complexity. In *Proc.*
492 *39th AAAI Conf. on Artificial Intelligence*, pp. 15347–15355, 2025.

493

494 Ibrahim M Almanjahie, Zouaoui Chikr Elmezouar, Ali Laksaci, and Mustapha Rachdi. KNN Local
495 Linear Estimation of the Conditional Cumulative Distribution Function: Dependent Functional
496 Data Case. *Comptes Rendus. Mathématique*, 356(10):1036–1039, 2018.

497

498 David Alvarez Melis and Tommi Jaakkola. Towards Robust Interpretability with Self-Explaining
499 Neural Networks. In *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.

500

501 Leila Amgoud and Martin Cooper. Axiomatic Foundations of Counterfactual Explanations. 2026.
502 Technical Report. <https://arxiv.org/abs/2602.04028>.

503

504 Guy Amir, Shahaf Bassan, and Guy Katz. Hard to Explain: On the Computational Hardness of
505 In-Distribution Model Interpretation. In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*,
506 2024.

507

508 Alaa Anani, Tobias Lorenz, Mario Fritz, and Bernt Schiele. Pixel-level Certified Explanations via
509 Randomized Smoothing. In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025.

510

511 Marcelo Arenas, D. Baez, P. Barceló, J. Pérez, and B. Subercaseaux. Foundations of Symbolic
512 Languages for Model Interpretability. In *Proc. 34th Int. Conf. on Advances in Neural Information*
513 *Processing Systems (NeurIPS)*, pp. 11690–11701, 2021.

514

515 Marcelo Arenas, P. Barceló, M. Romero, and B. Subercaseaux. On Computing Probabilistic Explana-
516 tions for Decision Trees. In *Proc. 35th Int. Conf. on Advances in Neural Information Processing*
517 *Systems (NeurIPS)*, pp. 28695–28707, 2022.

518

519 Marcelo Arenas, Pablo Barceló, Leopoldo Bertossi, and Mikaël Monet. On the Complexity of SHAP-
520 Score-Based Explanations: Tractability via Knowledge Compilation and Non-Approximability
521 Results. *Journal of Machine Learning Research (JMLR)*, 24(63):1–58, 2023.

522

523 Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre
524 Marquis. On Preferred Abductive Explanations for Decision Trees and Random Forests. In *Proc.*
525 *31st Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 643–650, 2022a.

526

527 Gilles Audemard, Steve Bellart, Louenas Bounia, Frédéric Koriche, Jean-Marie Lagniez, and Pierre
528 Marquis. Trading Complexity for Sparsity in Random Forest Explanations. In *Proc. 36th AAAI*
529 *Conf. on Artificial Intelligence*, number 5, pp. 5461–5469, 2022b.

530

531 Gilles Audemard, Jean-Marie Lagniez, Pierre Marquis, and Nicolas Szczepanski. Computing
532 Abductive Explanations for Boosted Trees. In *Proc. 26th Int. Conf. on Artificial Intelligence and*
533 *Statistics (AISTATS)*, pp. 4699–4711, 2023.

534

535 Steve Azzolin, Sagar Malhorta, Andrea Passerini, and Stefano Teso. Beyond Topological Self-
536 Explainable GNNs: A Formal Explainability Perspective. In *Proc. 42nd Int. Conf. on Machine*
537 *Learning (ICML)*, 2025.

538

539 Esma Balkir, Isar Nejadgholi, Kathleen C Fraser, and Svetlana Kiritchenko. Necessity and Sufficiency
for Explaining Text Classifiers: A Case Study in Hate Speech Detection. In *Proc. Conf. of the*
North American Chapter of the Association for Computational Linguistics: Human Language
Technologies (NAACL-HLT), pp. 2672–2686, 2022.

540 Pablo Barceló, M. Monet, J. Pérez, and B. Subercaseaux. Model Interpretability through the Lens of
541 Computational Complexity. In *Proc. 33rd Int. Conf. on Advances in Neural Information Processing*
542 *Systems (NeurIPS)*, pp. 15487–15498, 2020.

543 Pablo Barceló, Alexander Kozachinskiy, Miguel Romero, Bernardo Subercaseaux, and José Verschae.
544 Explaining k-Nearest Neighbors: Abductive and Counterfactual Explanations. In *Proc. 3rd ACM*
545 *on Management of Data*, pp. 1–26, 2025.

546 Shahaf Bassan and G. Katz. Towards Formal XAI: Formally Approximate Minimal Explanations
547 of Neural Networks. In *Proc. 29th Int. Conf. on Tools and Algorithms for the Construction and*
548 *Analysis of Systems (TACAS)*, pp. 187–207, 2023.

549 Shahaf Bassan, Guy Amir, Davide Corsi, Idan Refaeli, and Guy Katz. Formally Explaining Neural
550 Networks Within Reactive Systems. In *Proc. 23rd Int. Conf. on Formal Methods in Computer-Aided*
551 *Design (FMCAD)*, pp. 1–13, 2023.

552 Shahaf Bassan, Guy Amir, and Guy Katz. Local vs. Global Interpretability: A Computational
553 Complexity Perspective. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, 2024.

554 Shahaf Bassan, Guy Amir, Meirav Zehavi, and Guy Katz. What makes an Ensemble (Un) Inter-
555 pretable? In *Proc. 42nd Int. Conf. on Machine Learning (ICML)*, 2025a.

556 Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Matthias Althoff, and Guy Katz. Explaining,
557 Fast and Slow: Abstraction and Refinement of Provable Explanations. In *Proc. 42nd Int. Conf. on*
558 *Machine Learning (ICML)*, 2025b.

559 Shahaf Bassan, Ron Eliav, and Shlomit Gur. Explain Yourself, Briefly! Self-Explaining Neural
560 Networks with Concise Sufficient Reasons. In *Proc. 13th Int. Conf. on Learning Representations*
561 *(ICLR)*, 2025c.

562 Shahaf Bassan, Shlomit Gur, Sergey Zeltyn, Konstantinos Mavrogiorgos, Ron Eliav, and Dimosthenis
563 Kyriazis. Self-Explaining Neural Networks for Business Process Monitoring. 2025d. Technical
564 Report. <https://arxiv.org/abs/2503.18067>.

565 Shahaf Bassan, Michal Moshkovitz, and Guy Katz. Additive Models Explained: A Computational
566 Complexity Approach. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*,
567 2025e.

568 Shahaf Bassan, Yizhak Yisrael Elboher, Tobias Ladner, Volkan Şahin, Jan Kretinsky, Matthias Althoff,
569 and Guy Katz. Provably Explaining Neural Additive Models. In *Proc. 14th Int. Conf. on Learning*
570 *Representations (ICLR)*, 2026.

571 Robi Bhattacharjee and Ulrike Luxburg. Auditing Local Explanations is Hard. In *Proc. 38th Conf.*
572 *on Neural Information Processing Systems (NeurIPS)*, pp. 18593–18632, 2024.

573 Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility Theorems for Feature
574 Attribution. *Proc. of the National Academy of Sciences (PNAS)*, 121(2):e2304406120, 2024.

575 Guy Blanc, Jane Lange, and Li-Yang Tan. Provably Efficient, Succinct, and Precise Explanations.
576 In *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp.
577 6129–6141, 2021.

578 Guy Blanc, Caleb Koch, Jane Lange, and Li-Yang Tan. A Query-Optimal Algorithm for Finding
579 Counterfactuals. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 2075–2090, 2022.

580 Ryma Boumazouza, Fahima Cheikh-Alili, Bertrand Mazure, and Karim Tabia. ASTERYX: A model-
581 Agnostic SaT-basEd appRoach for SYmbolic and Score-Based EXplanations. In *Proc. 30th ACM*
582 *Int. Conf. on Information & Knowledge Management (CIKM)*, pp. 120–129, 2021.

583 Ryma Boumazouza, Raya Elsaleh, Melanie Ducoffe, Shahaf Bassan, and Guy Katz. FAME: For-
584 mal Abstract Minimal Explanation for Neural Networks. In *Proc. 14th Int. Conf. on Learning*
585 *Representations (ICLR)*, 2026.

594 Louenas Bounia. Enhancing the Intelligibility of Boolean Decision Trees with Concise and Reliable
595 Probabilistic Explanations. In *Proc. 20th Int. Conf. on Information Processing and Management of*
596 *Uncertainty in Knowledge-Based Systems (IPMU)*, pp. 205–218, 2024.

597 Louenas Bounia. Using Submodular Optimization to Approximate Minimum-Size Abductive Path
598 Explanations for Tree-Based Models. In *Proc. 41st Conf. on Uncertainty in Artificial Intelligence*
599 *(UAI)*, 2025.

600 Louenas Bounia and Frederic Koriche. Approximating Probabilistic Explanations via Supermodular
601 Minimization. In *Proc. 39th Conf. on Uncertainty in Artificial Intelligence (UAI)*, pp. 216–225,
602 2023.

603 Marco Calautti, Enrico Malizia, and Cristian Molinaro. On the Complexity of Global Necessary
604 Reasons to Explain Classification. 2025. Technical Report. [https://arXiv.org/abs/](https://arXiv.org/abs/2501.06766)
605 [2501.06766](https://arXiv.org/abs/2501.06766).

606 Emilio Carrizosa, Jasone Ramírez-Ayerbe, and Dolores Romero Morales. Mathematical Optimization
607 Modelling for Group Counterfactual Explanations. *European Journal of Operational Research*
608 *(EJOR)*, 319(2):399–412, 2024.

609 Brandon Carter, Jonas Mueller, Siddhartha Jain, and David Gifford. What Made You Do This?
610 Understanding Black-Box Decisions with Sufficient Input Subsets. In *Proc. 22nd Int. Conf. on*
611 *Artificial Intelligence and Statistics (AISTATS)*, pp. 567–576, 2019.

612 Brandon Carter, Siddhartha Jain, Jonas W Mueller, and David Gifford. Overinterpretation Reveals
613 Image Classification Model Pathologies. In *Proc. 35th Conf. on Neural Information Processing*
614 *Systems (NeurIPS)*, pp. 15395–15407, 2021.

615 Wenjing Chen and Victoria Crawford. Bicriteria Approximation Algorithms for the Submodular
616 Cover Problem. In *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp.
617 72705–72716, 2023.

618 Arthur Conmy, Augustine Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-
619 Alonso. Towards Automated Circuit Discovery for Mechanistic Interpretability. *Proc. 37th Conf.*
620 *on Neural Information Processing Systems (NeurIPS)*, pp. 16318–16352, 2023.

621 Yves Crama and Peter L Hammer. *Boolean Functions: Theory, Algorithms, and Applications*.
622 Cambridge University Press, 2011.

623 Adnan Darwiche. Logic for Explainable AI. In *Proc. 38th ACM/IEEE Symposium on Logic in*
624 *Computer Science (LICS)*, pp. 1–11, 2023.

625 Adnan Darwiche and Auguste Hirth. On the Reasons Behind Decisions. In *Proc. 24th European*
626 *Conf. on Artificial Intelligence (ECAI)*, pp. 712–720, 2020.

627 Adnan Darwiche and Chunxi Ji. On the Computation of Necessary and Sufficient Explanations. In
628 *Proc. 36th AAAI Conf. on Artificial Intelligence*, pp. 5582–5591, 2022.

629 Sanjoy Dasgupta, Nave Frost, and Michal Moshkovitz. Framework for Evaluating Faithfulness of
630 Local Explanations. In *Proc. 39th Int. Conf. on Machine Learning (ICML)*, pp. 4794–4815, 2022.

631 Ian Davidson, Nicolás Kennedy, and SS Ravi. CXAD: Contrastive Explanations for Anomaly
632 Detection: Algorithms, Complexity Results and Experiments. *Transactions on Machine Learning*
633 *Research (TMLR)*, 2025.

634 Alessandro De Palma, Greta Dolcetti, and Caterina Urban. Faster Verified Explanations for Neural
635 Networks. 2025. Technical Report. <https://arXiv.org/abs/2512.00164>.

636 Amit Dhurandhar, Pin-Yu Chen, Ronny Luss, Chun-Chen Tu, Paishun Ting, Karthikeyan Shanmugam,
637 and Payel Das. Explanations Based on the Missing: Towards Contrastive Explanations with
638 Pertinent Negatives. *Proc. 32nd Conf. on Neural Information Processing Systems (NeurIPS)*, 2018.

639 Hidde Fokkema, Rianne De Heide, and Tim Van Erven. Attribution-Based Explanations that Provide
640 Recourse Cannot be Robust. *Journal of Machine Learning Research (JMLR)*, 24(360):1–37, 2023.

-
- 648 Hidde Fokkema, Damien Garreau, and Tim van Erven. The Risks of Recourse in Binary Classification.
649 In *Proc. 27th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, pp. 550–558, 2024.
- 650
- 651 Tobias Geibinger, Reijo Jaakkola, Antti Kuusisto, Xinghan Liu, and Miikka Vilander. Why This and
652 Not That? A Logic-based Framework for Contrastive Explanations. In *Proc. 19th European Conf.
653 on Logics in Artificial Intelligence (JELIA)*, pp. 45–60. Springer, 2025.
- 654
- 655 Joshua Größer and Ostap Okhrin. Copulae: An Overview and Recent Developments. *Wiley Interdis-*
656 *ciplinary Reviews: Computational Statistics*, 14(3):e1557, 2022.
- 657
- 658 Riccardo Guidotti. Counterfactual Explanations and How to Find Them: Literature Review and
659 Benchmarking. *Data Mining and Knowledge Discovery (DMKD)*, 38(5):2770–2824, 2024.
- 660
- 661 Hyukjun Gweon, Matthias Schonlau, and Stefan H Steiner. The K Conditional Nearest Neighbor
662 Algorithm for Classification and Class Probability Estimation. *PeerJ Computer Science*, 5:e194,
2019.
- 663
- 664 Itamar Hadad, Shahaf Bassan, and Guy Katz. Formal Mechanistic Interpretability: Automated Circuit
665 Discovery with Provable Guarantees. In *Proc. 14th Int. Conf. on Learning Representations (ICLR)*,
2026.
- 666
- 667 Ouns El Harzli, Bernardo Cuenca Grau, and Ian Horrocks. Cardinality-Minimal Explanations for
668 Monotonic Neural Networks. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp.
669 3677–3685, 2023.
- 670
- 671 W Keith Hastings. Monte Carlo Sampling Methods using Markov Chains and their Applications.
672 *Biometrika*, 57:97–109, 1970.
- 673
- 674 Rongjie Huang, Max W. Y. Lam, Jun Wang, Dan Su, Dong Yu, Yi Ren, and Zhou Zhao. FastDiff: A
675 Fast Conditional Diffusion Model for High-Quality Speech Synthesis. In *Proc. 31st Int. Joint Conf.
on Artificial Intelligence (IJCAI)*, pp. 4157–4163, 2022.
- 676
- 677 Xuanxiang Huang, Yacine Izza, Alexey Ignatiev, and Joao Marques-Silva. On Efficiently Explaining
678 Graph-Based Classifiers. In *Proc. 18th Int. Conf. on Principles of Knowledge Representation and
Reasoning (KR)*, 2021.
- 679
- 680 Alexey Ignatiev, Nina Narodytska, and Joao Marques-Silva. Abduction-Based Explanations for
681 Machine Learning Models. In *Proc. 33rd AAAI Conference on Artificial Intelligence*, pp. 1511–
682 1519, 2019.
- 683
- 684 Alexey Ignatiev, Martin Cooper, Mohamed Siala, Emmanuel Hebrard, and Joao Marques-Silva.
685 Towards Formal Fairness in Machine Learning. In *Proc. 26th Int. Conf. on Principles and Practice
of Constraint Programming (CP)*, pp. 846–867, 2020a.
- 686
- 687 Alexey Ignatiev, Nina Narodytska, Nicholas Asher, and Joao Marques-Silva. From Contrastive to
688 Abductive Explanations and Back Again. In *Proc. Int. Conf. of the Italian Association for Artificial
689 Intelligence (AIxAI)*, pp. 335–355, 2020b.
- 690
- 691 Alexey Ignatiev, Yacine Izza, Peter J Stuckey, and Joao Marques-Silva. Using MaxSAT for Efficient
692 Explanations of Tree Ensembles. In *Proc. 36th AAAI Conference on Artificial Intelligence*, pp.
3776–3785, 2022.
- 693
- 694 Rishabh K Iyer and Jeff A Bilmes. Submodular Optimization with Submodular Cover and Submodular
695 Knapsack Constraints. *Proc. 27th Conf. on Neural Information Processing Systems (NeurIPS)*,
2013.
- 696
- 697 Yacine Izza and Joao Marques-Silva. On Explaining Random Forests with SAT. In *Proc. 30th Int.
698 Joint Conf. on Artificial Intelligence (IJCAI)*, 2021.
- 699
- 700 Yacine Izza, Xuanxiang Huang, Alexey Ignatiev, Nina Narodytska, Martin Cooper, and Joao Marques-
701 Silva. On Computing Probabilistic Abductive Explanations. *Int. Journal of Approximate Reasoning
(IJAR)*, 159:108939, 2023.

-
- 702 Yacine Izza, Xuanxiang Huang, Antonio Morgado, Jordi Planes, Alexey Ignatiev, and Joao Marques-
703 Silva. Distance-Restricted Explanations: Theoretical Underpinnings & Efficient implementation.
704 In *Proc. 21st Int. Conf. on Principles of Knowledge Representation and Reasoning (KR)*, pp.
705 475–486, 2024a.
- 706 Yacine Izza, Kuldeep S Meel, and Joao Marques-Silva. Locally-Minimal Probabilistic Explanations.
707 In *Proc. 27th European Conf. on Artificial Intelligence (ECAI)*, 2024b.
- 708 Reijo Jaakkola, Tomi Janhunen, Antti Kuusisto, Masood Feyzbakhsh Rankooh, and Miikka Vilander.
709 Explainability via Short Formulas: the Case of Propositional Logic with Implementation. *Journal*
710 *of Artificial Intelligence Research (JAIR)*, 83, 2025.
- 711 Alon Jacovi, Swabha Swayamdipta, Shauli Ravfogel, Yanai Elazar, Yejin Choi, and Yoav Goldberg.
712 Contrastive Explanations for Model Interpretability. In *Proc. Conf. Conference on Empirical*
713 *Methods in Natural Language Processing (EMNLP)*, pp. 1597–1611, 2021.
- 714 Helen Jin, Anton Xue, Weiqiu You, Surbhi Goel, and Eric Wong. Probabilistic Stability Guarantees
715 for Feature Attributions. In *Proc. 39th Conf. on Neural Information Processing Systems (NeurIPS)*,
716 2025.
- 717 Amir-Hossein Karimi, Bernhard Schölkopf, and Isabel Valera. Algorithmic recourse: from counter-
718 factual explanations to interventions. In *Proc. 4th ACM Conf. on Fairness, Accountability, and*
719 *Transparency (FAccT)*, pp. 353–362, 2021.
- 720 Eoin M Kenny and Mark T Keane. On Generating Plausible Counterfactual and Semi-Factual
721 Explanations for Deep Learning. In *Proc. 35th AAAI Conf. on Artificial Intelligence*, number 13,
722 pp. 11575–11585, 2021.
- 723 Alexander Kozachinskiy. Inapproximability of Sufficient Reasons for Decision Trees. 2023. Technical
724 Report. <https://arXiv.org/abs/2304.02781>.
- 725 I Elizabeth Kumar, Suresh Venkatasubramanian, Carlos Scheidegger, and Sorelle Friedler. Problems
726 with Shapley-Value-Based Explanations as Feature Importance Measures. In *Proc. 37th Int. Conf.*
727 *on Machine Learning (ICML)*, pp. 5491–5500, 2020.
- 728 Emanuele La Malfa, Agnieszka Zbrzezny, Rhiannon Michelmore, Nicola Paoletti, and Marta
729 Kwiatkowska. On Guaranteed Optimal Robust Explanations for NLP Models. In *Proc. Int.*
730 *Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 2658–2665, 2021.
- 731 Eduardo Sany Laber. The Computational Complexity of some Explainable Clustering Problems.
732 *Information Processing Letters*, 184:106437, 2024.
- 733 Shuai Li, Yingjie Zhang, Hongtu Zhu, Christina Wang, Hai Shu, Ziqi Chen, Zhuoran Sun, and
734 Yanfeng Yang. K-Nearest-Neighbor Local Sampling Based Conditional Independence Testing.
735 *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp. 23321–23344, 2023.
- 736 Gianluigi Lopardo, Frédéric Precioso, and Damien Garreau. A Sea of Words: An In-Depth Analysis
737 of Anchors for Text Data. In *Proc. 26th Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*
738 2023, 2023.
- 739 Scott M Lundberg and Su-In Lee. A Unified Approach to Interpreting Model Predictions. *Proc. 31st*
740 *Conf. on Neural Information Processing Systems (NeurIPS)*, 2017.
- 741 Joao Marques-Silva and Alexey Ignatiev. Delivering Trustworthy AI through formal XAI. In *Proc.*
742 *36th AAAI Conf. on Artificial Intelligence*, pp. 12342–12350, 2022.
- 743 Joao Marques-Silva and Alexey Ignatiev. No Silver Bullet: Interpretable ML Models Must be
744 Explained. *Frontiers in Artificial Intelligence*, 6:1128212, 2023.
- 745 Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska.
746 Explaining Naive Bayes and Other Linear Classifiers with Polynomial Time and Delay. In *Proc.*
747 *33rd Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp. 20590–
748 20600, 2020.

756 Joao Marques-Silva, Thomas Gerspacher, Martin Cooper, Alexey Ignatiev, and Nina Narodytska.
757 Explanations for Monotonic Classifiers. In *Proc. Int. Conf. on Machine Learning (ICML)*, pp.
758 7469–7479, 2021.

759 Reda Marzouk and Colin De La Higuera. On the Tractability of SHAP Explanations under Markovian
760 Distributions. In *Proc. 41st Int. Conf. on Machine Learning (ICML)*, pp. 34961–34986, 2024.

761 Reda Marzouk, Shahaf Bassan, and Guy Katz. SHAP Meets Tensor Networks: Provably Tractable
762 Explanations with Parallelism. In *Proc. 39th Conf. on Neural Information Processing Systems*
763 *(NeurIPS)*, 2025a.

764 Reda Marzouk, Shahaf Bassan, Guy Katz, and De la Higuera. On the Computational Tractability
765 of the (Many) Shapley Values. In *Proc. 28th Int. Conf. on Artificial Intelligence and Statistics*
766 *(AISTATS)*, 2025b.

767 Kuldeep S Meel, Aditya A Shrotri, and Moshe Y Vardi. Not all FPRASs are Equal: Demystifying
768 FPRASs for DNF-Counting. *Constraints*, 24(3):211–233, 2019.

769 Ramaravind K Mothilal, Amit Sharma, and Chenhao Tan. Explaining Machine Learning Classifiers
770 Through Diverse Counterfactual Explanations. In *Proc. 3rd Conf. on Fairness, Accountability, and*
771 *Transparency (FAccT)*, pp. 607–617, 2020.

772 Vidya Muthukumar, Tejaswini Pedapati, Nalini Ratha, Prasanna Sattigeri, Chai-Wah Wu, Brian
773 Kingsbury, Abhishek Kumar, Samuel Thomas, Aleksandra Mojsilovic, and Kush R Varshney.
774 Understanding Unequal Gender Classification Accuracy from Face Images. 2018. Technical
775 Report. <https://arXiv.org/abs/1812.00099>.

776 Sebastian Ordyniak, Giacomo Paesani, and Stefan Szeider. The Parameterized Complexity of Finding
777 Concise Local Explanations. In *Proc. 32nd Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2023.

778 Sebastian Ordyniak, Giacomo Paesani, Mateusz Rychlicki, and Stefan Szeider. Explaining Decisions
779 in ML Models: A Parameterized Complexity Analysis. In *Proc. 21st Int. Conf. on Principles of*
780 *Knowledge Representation and Reasoning (KR)*, pp. 563–573, 2024.

781 Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. Anchors: High-Precision Model-Agnostic
782 Explanations. In *Proc. 32nd AAAI Conf. on Artificial Intelligence*, 2018.

783 Majun Shi, Zishen Yang, and Wei Wang. Minimum Non-Submodular Cover Problem with Applica-
784 tions. *Applied Mathematics and Computation*, 410:126442, 2021.

785 Andy Shih, Arthur Choi, and Adnan Darwiche. A Symbolic Approach to Explaining Bayesian
786 Network Classifiers. In *Proc. 27th Int. Joint Conf. on Artificial Intelligence (IJCAI)*, pp. 5103–
787 5111, 2018.

788 Bernardo Subercaseaux, Marcelo Arenas, and Kuldeep S Meel. Probabilistic Explanations for Linear
789 Models. In *Proc. 39th AAAI Conf. on Artificial Intelligence*, pp. 20655–20662, 2025.

790 Guy Van den Broeck, Anton Lykov, Maximilian Schleich, and Dan Suciu. On the Tractability of
791 SHAP Explanations. *Journal of Artificial Intelligence Research (JAIR)*, 74:851–886, 2022.

792 Gal Vardi, Gilad Yehudai, and Ohad Shamir. On the Optimal Memorization Power of ReLU Neural
793 Networks. In *Proc. 10th Int. Conf. on Learning Representations, (ICLR)*, 2022.

794 Vijay V Vazirani. *Approximation Algorithms*. Springer, 2001.

795 Sahil Verma, Varich Boonsanong, Minh Hoang, Keegan Hines, John Dickerson, and Chirag Shah.
796 Counterfactual Explanations and Algorithmic Recourses for Machine Learning: A Review. *ACM*
797 *Computing Surveys*, 56(12):1–42, 2024.

798 Stephan Wäldchen, Jan Macdonald, Sascha Hauch, and Gitta Kutyniok. The Computational Com-
799 plexity of Understanding Binary Classifier Decisions. *Journal of Artificial Intelligence Research*
800 *(JAIR)*, 70:351–387, 2021.

801 E. Wang, P. Khosravi, and G. Van den Broeck. Probabilistic Sufficient Explanations. In *Proc. 30th*
802 *Int. Joint Conf. on Artificial Intelligence (IJCAI)*, 2021a.

-
- 810 S. Wang, H. Zhang, K. Xu, X. Lin, S. Jana, C. Hsieh, and Z. Kolter. Beta-Crown: Efficient Bound
811 Propagation with Per-Neuron Split Constraints for Neural Network Robustness Verification. In
812 *Proc. 35th Int. Conf. on Advances in Neural Information Processing Systems (NeurIPS)*, pp.
813 29909–29921, 2021b.
- 814 Greta Warren, Eoin Delaney, Christophe Guéret, and Mark T Keane. Explaining Multiple Instances
815 Counterfactually: User Tests of Group-Counterfactuals for XAI. In *Proc. 32nd Int. Conf. on*
816 *Case-Based Reasoning (ICCBR)*, pp. 206–222, 2024.
- 817 Ingo Wegener. The Complexity of Symmetric Boolean Functions. *Computation Theory and Logic*,
818 pp. 433–442, 2005.
- 819 Laurence A Wolsey. An Analysis of the Greedy Algorithm for the Submodular Set Covering Problem.
820 *Combinatorica*, 2(4):385–393, 1982.
- 821 Haoze Wu, Omri Isac, Aleksandar Zeljić, Teruhiro Tagomori, Matthew Daggitt, Wen Kokke, Idan
822 Refaeli, Guy Amir, Kyle Julian, Shahaf Bassan, et al. Marabou 2.0: A Versatile Formal Analyzer
823 of Neural Networks. In *Proc. 36th Int. Conf. on Computer Aided Verification (CAV)*, pp. 249–264,
824 2024a.
- 825 Min Wu, Haoze Wu, and Clark Barrett. Verix: Towards Verified Explainability of Deep Neural
826 Networks. In *Proc. 36th Int. Conf. on Advances in Neural Information Processing Systems*
827 *(NeurIPS)*, 2024b.
- 828 Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling Tabular
829 Data Using Conditional GAN. *Proc. 33rd Conf. on Neural Information Processing Systems*
830 *(NeurIPS)*, 2019.
- 831 Anton Xue, Rajeev Alur, and Eric Wong. Stability Guarantees for Feature Attributions with Multi-
832 plicative Smoothing. *Proc. 37th Conf. on Neural Information Processing Systems (NeurIPS)*, pp.
833 62388–62413, 2023.
- 834 Kayo Yin and Graham Neubig. Interpreting Language Models with Contrastive Explanations. In
835 *Proc. Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 184–198, 2022.
- 836 Jinqiang Yu, Alexey Ignatiev, Peter Stuckey, Nina Narodytska, and Joao Marques-Silva. Eliminating
837 the Impossible, Whatever Remains Must Be True: On Extracting and Applying Background
838 Knowledge in the Context of Formal Explanations. In *Proc. 37th AAAI Conf. on Artificial*
839 *Intelligence*, pp. 4123–4131, 2023.
- 840
841
842
843
844
845
846
847
848
849
850
851
852
853
854
855
856
857
858
859
860
861
862
863

Appendix

The appendix contains formalizations and proofs that were mentioned throughout the paper:

Appendix A contains extended related work on Formal XAI.

Appendix B contains the formalizations of the models and distributions used in this work.

Appendix C contains the proof of Proposition 1.

Appendix D contains the proof of Proposition 2.

Appendix E contains the proof of Proposition 3.

Appendix F contains the proof of Proposition 4.

Appendix G contains the proof of Proposition 5.

Appendix H contains the proof of Theorem 1.

Appendix I contains the proof of Proposition 6.

Appendix J contains the proof of Theorem 2.

Appendix K contains the proof of Theorem 3.

Appendix L contains the proof of Theorem 4.

Appendix O contains an LLM usage disclosure.

A EXTENDED RELATED WORK ON FORMAL XAI

Our work contributes to *formal explainable AI* (formal XAI) (Marques-Silva & Ignatiev, 2022), the line of work that studies explanations with provable guarantees (Yu et al., 2023; Darwiche & Ji, 2022; Darwiche, 2023; Shih et al., 2018; Azzolin et al., 2025; Audemard et al., 2021; Calautti et al., 2025). The most relevant thread for this paper is the computational complexity of producing such guarantees (Barceló et al., 2020; ?; Marzouk et al., 2025a;b; Marzouk & De La Higuera, 2024; Bassan et al., 2025e; Amir et al., 2024; Blanc et al., 2021; 2022; Jaakkola et al., 2025; Amgoud & Cooper, 2026; Geibinger et al., 2025). Existing results often show intractability for expressive models such as neural networks (Adolfi et al., 2024; Barceló et al., 2020) and tree ensembles (Ordyniak et al., 2024; Bassan et al., 2025a), which motivates the literature’s emphasis on tractable model classes (Marques-Silva & Ignatiev, 2023), including decision trees (Bounia, 2025; Arenas et al., 2022; Bounia, 2024; Bounia & Koriche, 2023), monotonic models (Marques-Silva et al., 2021; Harzli et al., 2023), linear models (Marques-Silva et al., 2020; Subercaseaux et al., 2025), and additive models (Bassan et al., 2026). Our contribution complements this map by showing that global explanation objectives can have exploitable structure even when local counterparts are hard.

A second related thread develops practical solvers for guaranteed explanations. This includes MaxSAT and MILP encodings for tree ensembles (Audemard et al., 2023; 2022a; Ignatiev et al., 2022; Boumazouza et al., 2021), as well as neural-network verification frameworks (Wang et al., 2021b; Wu et al., 2024a) used to obtain certified explanations for neural networks (Wu et al., 2024b; Bassan & Katz, 2023; Bassan et al., 2023; Izza et al., 2024a; Hadad et al., 2026; Ignatiev et al., 2019). Other work reduces the computational burden through abstractions (De Palma et al., 2025; Bassan et al., 2025b; Boumazouza et al., 2026), relaxations such as smoothing (Xue et al., 2023; Anani et al., 2025; Jin et al., 2025), or training-time interventions (Alvarez Melis & Jaakkola, 2018; Bassan et al., 2025c;d). Our paper is orthogonal to these implementations: it identifies when the objective itself has monotone, submodular, or supermodular structure.

Finally, within formal XAI, a related line of work studies methods that leverage combinatorial and submodular optimization techniques to improve the computation of *local* probabilistic sufficient reasons (Bounia, 2025; Bounia & Koriche, 2023), particularly for decision tree classifiers. As noted in that literature, as well as in this work, the value-function objective underlying these local explanation variants is not submodular in practice. Nevertheless, greedy algorithms inspired by submodular optimization often serve as effective heuristics and can yield high-quality explanations empirically. In contrast, our work establishes a rigorous theoretical connection between combinatorial and submodular optimization objectives and the *global* variants of sufficient and contrastive reasons. This connection provides formal guarantees and opens the door to a broader range of principled algorithmic implementations.

B MODEL AND DISTRIBUTION FORMALIZATIONS

This appendix records the model and distribution formalizations used by the contribution statements in the main text. Subsection B.1 defines the model families used in the complexity results, and Subsection B.2 defines the distribution classes that determine which value-function properties apply.

B.1 MODEL FORMALIZATIONS

We formalize the model classes that instantiate the paper’s complexity separations: (i) (axis-aligned) decision trees, (ii) linear classifiers, and (iii) neural networks with ReLU activations.

Decision Trees. Decision trees provide the interpretable baseline for our local/global separation. We define a decision tree (DT) as a directed acyclic graph representing $f : \mathbb{F} \rightarrow [c]$, where $c \in \mathbb{N}$ is the number of classes. The graph encodes the function as follows: (i) Each internal node v is assigned a distinct binary input feature from the set $\{1, \dots, n\}$; (ii) Every internal node v has at most k outgoing edges, each corresponding to a value in $[k]$ assigned to v ; (iii) Along any path α in the decision tree, each variable appears at most once; (iv) Each leaf node is labeled with a class from $[c]$. An input $\mathbf{x} \in \mathbb{F}$ therefore determines a unique root-to-leaf path α , and $f(\mathbf{x})$ is the class label at the reached leaf. The size $|f|$ is the total number of graph edges. Different paths α and α' may use different variable orders, but no variable repeats along a single path.

Neural Networks. We present our *hardness* proofs for neural networks with ReLU activations, though our *tractability results* (i.e., polynomial-time algorithms) apply to any architecture that allows for polynomial-time inference — a standard assumption. Thus, all separation results between tractable and intractable cases we prove carry over to *any neural architecture* at least as expressive as a standard feed-forward ReLU network, encompassing many widely used models such as ResNets, CNNs, Transformers, Diffusion models, and more. Moreover, note that any ReLU network can be represented as a fully connected network by assigning zero weights and biases to missing connections. Following standard conventions (Barceló et al., 2020; Bassan et al., 2024; Adolphi et al., 2025), we thus assume the network is fully connected. Specifically, our analysis applies to multi-layer perceptrons (MLPs). Formally, an MLP f consists of $t - 1$ *hidden layers* ($g^{(j)}$ for $j = 1, \dots, t - 1$) and one output layer $g^{(t)}$, where each layer is defined recursively as:

$$g^{(j)} := \sigma^{(j)}(g^{(j-1)}W^{(j)} + b^{(j)}) \quad (9)$$

where $W^{(j)}$ denotes the weight matrix, $b^{(j)}$ the bias vector, and $\sigma^{(j)}$ the activation function of the j -th layer. Accordingly, the model comprises t weight matrices ($W^{(1)}, \dots, W^{(t)}$), t bias vectors ($b^{(1)}, \dots, b^{(t)}$), and t activation functions ($\sigma^{(1)}, \dots, \sigma^{(t)}$).

The input layer is $g^{(0)} = \mathbf{x} \in \{0, 1\}^n$. A sequence of positive integers $d_0, \dots, d_t$ fixes the layer dimensions, with rational weights and biases $W^{(j)} \in \mathbb{Q}^{d_{j-1} \times d_j}$ and $b^{(j)} \in \mathbb{Q}^{d_j}$. For the binary classifiers used in the hardness and tractability statements, $d_0 = n$ and $d_t = 1$. Hidden layers use the ReLU activation $\text{ReLU}(x) = \max(0, x)$. Although sigmoid activations are typical during training, the post-hoc explanation setting lets us use a threshold output layer, with $\text{step}(\mathbf{z}) = 1$ if $\mathbf{z} \geq 0$ and $\text{step}(\mathbf{z}) = 0$ otherwise.

Tree Ensembles. Tree ensembles are included to show that the global tractability results extend beyond single transparent models. Because the explanations are post-hoc, we formalize the *inference* phase rather than training. The main ensemble families use either *majority voting* (e.g., Random Forests) or *weighted voting* (e.g., XGBoost). As with neural networks, the polynomial-time tractability results apply to any ensemble configuration with polynomial-time inference.

Majority Voting Inference. For *majority voting*, the ensemble prediction $f(\mathbf{x})$ is the class $j \in [c]$ receiving the most votes among the individual tree predictions $f_i(\mathbf{x})$ for $i \in [k]$.

$$f(\mathbf{x}) := \text{majority}(\{f_i(\mathbf{x}) \mid i \in [k]\}) \quad (10)$$

Here, $\text{majority}(S)$ denotes the most frequent label in the multiset S , with ties resolved by a fixed rule such as lexicographic order or a predefined priority.

Weighted Voting Inference. For *weighted voting*, each base model receives a rational weight $\phi_i \in \mathbb{Q}$, and the predicted class is the one with the highest total supporting weight. Formally, for $\mathbf{x} \in \mathbb{F}$:

$$f(\mathbf{x}) := \arg \max_{j \in [c]} \sum_{i=1}^k \phi_i \cdot \mathbb{I}[f_i(\mathbf{x}) = j] \quad (11)$$

where $\mathbb{I}$ denotes the identity function.

B.2 DISTRIBUTION FORMALIZATIONS

This subsection formalizes the distribution classes that appear in the main contribution statements.

Empirical Distributions. Empirical distributions support the paper’s black-box tractability results because value terms can be computed by iterating over a finite dataset. Given $\mathbf{D} = \{\mathbf{z}^1, \mathbf{z}^2, \dots, \mathbf{z}^{|\mathbf{D}|}\}$, the probability of $\mathbf{x} \in \mathbb{F}$ is its frequency in $\mathbf{D}$:

$$\Pr(\mathbf{x}) := \frac{1}{|\mathbf{D}|} \sum_{i=1}^{|\mathbf{D}|} \mathbb{I}(\mathbf{z}^i = \mathbf{x}) \quad (12)$$

Independent Distributions. Independent distributions are the assumption under which the global sufficient and contrastive value functions obtain supermodularity and submodularity. Given feature probabilities $p(\mathbf{x}_i) \in [0, 1]$, $\mathcal{D}$ is independent when $\Pr(\mathbf{x}) := \prod_{i \in [n]} p(\mathbf{x}_i)$.

When $p(\mathbf{x}_i) = p(\mathbf{x}_j)$ for all $i, j \in [n]$, this specializes to the *uniform distribution*, which is therefore a special case of independence.

General Distributions. General distributions are needed for the value-function properties that hold without structural distributional assumptions. In practice, empirical distributions based on training data are common in XAI, but other distributional approximations include k-NN resampling from nearby points (Li et al., 2023; Almanjahie et al., 2018; Gweon et al., 2019), copulas for tabular dependencies (Größer & Okhrin, 2022), and conditional generative models such as CTGAN (Xu et al., 2019) or conditional diffusion models (Huang et al., 2022). These choices also appear in counterfactual explanation and algorithmic recourse work (Karimi et al., 2021; Fokkema et al., 2024; 2023; Verma et al., 2024), which is closely related to contrastive explanations. When the value function is not computed from model structure, such as leaf enumeration in decision trees, or from an empirical distribution, it can instead be approximated with methods such as Monte Carlo sampling (Hastings, 1970), following approaches similar to (Subercaseaux et al., 2025; Lopardo et al., 2023).

B.3 SUBSET VS. CARDINAL MINIMALITY

This subsection expands the minimality distinction used throughout the paper. Cardinal minimality is the stronger guarantee because it requires globally minimum explanation size; subset minimality only rules out explanations obtained by deleting features from the chosen set.

The implication is immediate: if $S \subseteq [n]$ is cardinally minimal, then no set S' with $|S'| < |S|$ is an explanation. In particular, no strict subset $S' \subsetneq S$ is an explanation, so S is subset minimal.

The converse fails, which is why the paper treats cardinal-minimal approximation as a strictly stronger contribution. Consider:

$$f := \mathbf{x}_1 \vee (\mathbf{x}_2 \wedge \mathbf{x}_3) \quad (13)$$

and the assignment $\mathbf{x} := (1, 1, 1)$, which gives $f(1, 1, 1) = 1$. Fixing feature $\mathbf{x}_1 = 1$ yields a cardinally minimal (and subset minimal) sufficient reason, since the prediction remains 1 regardless of $\mathbf{x}_2, \mathbf{x}_3$. But fixing both $\mathbf{x}_2 = 1$ and $\mathbf{x}_3 = 1$ also gives a subset minimal explanation—yet not a cardinally minimal one. Thus, while every cardinally minimal explanation is subset minimal, not every subset minimal explanation is cardinally minimal.

C PROOF OF PROPOSITION 1

Proposition 1. *While the non-probabilistic case ($\delta = 1$) admits a unique subset-minimal global sufficient reason (Bassan et al., 2024), in the general probabilistic setting (for arbitrary δ), there exist functions f that have $\Theta(\frac{2^n}{\sqrt{n}})$ subset-minimal global sufficient reasons.*

Proof. This proof establishes the non-uniqueness contribution announced in the main text: once global sufficiency is relaxed from $\delta = 1$ to arbitrary probabilistic thresholds, subset-minimal global sufficient reasons need not be unique and can appear in exponentially many forms.

It is known that certain Boolean functions admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons for some input $\mathbf{x}$ Bassan et al. (2024). In particular, this phenomenon was shown to occur in functions of the following form. We demonstrate that the same function admits an exponential number of *global* and *probabilistic* sufficient reasons. Notably, the mentioned function is a special case of a broader class of *threshold Boolean functions* described in Wegener (2005). Specifically, for $n = 2k + 1$ with $k \in \mathbb{N}$, this function is defined as:

$$f(\mathbf{x}) := \begin{cases} 1 & \text{if } \sum_{i=1}^n \mathbf{x}_i \geq k + 1 \\ 0 & \text{otherwise} \end{cases} \quad (14)$$

where $\mathbf{x}$ is drawn from a uniform distribution $\mathcal{D}$. Notably, the function f is *symmetric*, in the sense of symmetric threshold Boolean functions Wegener (2005): its output depends solely on the number of 1's (or, equivalently, 0's) in the input and is invariant under any permutation of input bits. Furthermore, the two extreme inputs — the all-zeros vector $(0, \dots, 0)$ and the all-ones vector $(1, \dots, 1)$ — each admit an exponential number of subset-minimal local (non-probabilistic) sufficient reasons.

Since the function f is symmetric, the condition $v_{\text{suff}}^g(S_1) = v_{\text{suff}}^g(S_2)$ holds for any two subsets $S_1, S_2 \subseteq [n]$. Moreover, it can be verified that for any subset $S \subseteq [n]$ of size $k + 1$, we have:

$$(\forall i \in S). \quad v_{\text{suff}}^g(S) > v_{\text{suff}}^g(S \setminus \{i\}) \quad (15)$$

We also deliberately set δ to satisfy:

$$v_{\text{suff}}^g(S \setminus \{i\}) < \delta \leq v_{\text{suff}}^g(S), \quad (16)$$

and also set $k := \lfloor \frac{n}{2} \rfloor$. Each of these subsets is *subset-minimal* because any subset of size at most $\lfloor \frac{n}{2} \rfloor - 1$ fails to be a sufficient reason for $\langle f, \mathbf{x} \rangle$ (with respect to the δ threshold). Therefore, we can directly apply the same analysis as in (Bassan et al., 2024). In particular, there are exactly $\binom{n}{\lfloor \frac{n}{2} \rfloor}$ subset-minimal local sufficient reasons for $\langle f, \mathbf{x} \rangle$. Using Stirling's approximation, we obtain:

$$\lim_{n \rightarrow \infty} \frac{2\sqrt{2\pi}}{e^2} \cdot \frac{2^n}{\sqrt{n}} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor} \leq \frac{e}{\pi} \cdot \frac{2^n}{\sqrt{n}} \quad (17)$$

This yields the corresponding bound on the number of subset-minimal global δ -sufficient reasons.

□

D PROOF OF PROPOSITION 2

Proposition 2. *While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.*

Proof. This proof supplies the structural monotonicity contribution used by the main complexity results. We prove monotonicity for both global sufficient and global contrastive explanations. The argument starts with global sufficiency for Boolean functions, then extends to discrete multi-valued

domains with multiple output classes, and then to continuous domains. Finally, we transfer the same monotonicity guarantee to global contrastive explanations. No step uses feature independence, so the monotonicity property holds for *any* underlying distribution.

We close the proof by contrasting this global result with a simple local counterexample, showing that the corresponding local probabilistic sufficient and contrastive value functions are not monotone.

Lemma 1. *The global sufficient value function v_{suff}^g is monotonic non-decreasing for Boolean functions, under any data distribution.*

Proof. Given an arbitrary set $S \in [n]$, and a feature $i \notin S$, we have that:

$$\Pr(\mathbf{x}) = \Pr(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) = \Pr(\mathbf{x}_{\bar{S}} \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \quad (18)$$

Consequently, this implies that:

$$\begin{aligned} \Pr(\mathbf{x}) &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_{S \cup \{i\}}) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(\mathbf{x}_S, x_i) \\ &= \Pr(\mathbf{x}_{\overline{S \cup \{i\}}} \mid \mathbf{x}_{S \cup \{i\}}) \cdot \Pr(x_i \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S) \end{aligned} \quad (19)$$

Moreover, for any two points $\mathbf{x}, \mathbf{x}' \sim \mathcal{D}$ such that $\mathbf{x}_S = \mathbf{x}'_S$ and $f(\mathbf{x}) = 1 - f(\mathbf{x}')$, it holds that:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) = 1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}') \mid \mathbf{z}_S = \mathbf{x}'_S) \quad (20)$$

To simplify notation, we occasionally omit the distribution notation $\sim \mathcal{D}$ in the local probability expression $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, and use f^+ and f^- to denote the events $f(\mathbf{z}) = 1$ and $f(\mathbf{z}) = 0$, respectively.

The proof first rewrites the probability term into a form where the monotonicity contribution is visible, and then uses that form to prove the claim.

Simplifying the probability term. Let $\mathcal{D}_S$ denote the distribution restricted to the set S . Given a point $\mathbf{x} \sim \mathcal{D}$ and a set S , consider all points $\mathbf{z} \sim \mathcal{D}$ such that $\mathbf{z}_S = \mathbf{x}_S$ and $f(\mathbf{z}) = b$, where $b \in \{0, 1\}$ — they all share the same local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S)$. For every $\mathbf{x} \sim \mathcal{D}$ that shares the same x_S and the same output b , we marginalize over $x_{\bar{S}}$. This yields $\Pr(f(\mathbf{x}) = b, \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S) \cdot \Pr(\mathbf{x}_S)$, from which we can infer:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = b \mid \mathbf{z}_S = \mathbf{x}_S) = \Pr(f(\mathbf{x}) = b \mid \mathbf{x}_S). \quad (21)$$

Thus, for $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$, we get that:

$$\begin{aligned} &\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^+} \Pr(f^+, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S) + \sum_{\mathbf{x}_S \sim \mathcal{D}_S, f^-} \Pr(f^-, \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S) \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (\Pr_{\mathbf{z} \sim \mathcal{D}}(f^- \mid \mathbf{z}_S = \mathbf{x}_S))^2] \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(\Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2 + (1 - \Pr_{\mathbf{z} \sim \mathcal{D}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S))^2], \end{aligned} \quad (22)$$

From which it follows that:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\mathbf{x}_{S \cup \{t\}} \sim \mathcal{D}_{S \cup \{t\}}} \Pr(\mathbf{x}_{S \cup \{t\}}) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(x_t = 1, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1))^2] + \\
&\quad \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(x_t = 0, \mathbf{x}_S) \cdot [(\Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2 + (1 - \Pr_{\mathbf{z}}(f^+ \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0))^2]
\end{aligned} \tag{23}$$

Proving monotonicity. We now proceed to prove the monotonicity claim, building on the previous simplification. We begin by introducing a few additional notations. Let $g_1^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 1)$ and $g_0^+ := \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_t = 0)$. Similarly, define $P_1 := \Pr(z_t = 1 \mid \mathbf{x}_S)$ and $P_0 := \Pr(z_t = 0 \mid \mathbf{x}_S)$. Then, the following expectations $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ and $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})]$ can be simplified accordingly¹:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 + (1 - (P_1 \cdot g_1^+ + P_0 \cdot g_0^+))^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2 - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{24}$$

As a result, we obtain the following:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot ((g_1^+)^2 + (1 - g_1^+)^2) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot ((g_0^+)^2 + (1 - g_0^+)^2) \\
&= \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=1}} \Pr(\mathbf{x}_S) \cdot P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + \sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t=0}} \Pr(\mathbf{x}_S) \cdot P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [P_1 \cdot (2(g_1^+)^2 - 2g_1^+ + 1) + P_0 \cdot (2(g_0^+)^2 - 2g_0^+ + 1)] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2P_1 \cdot ((g_1^+)^2 - g_1^+) + 2P_0 \cdot ((g_0^+)^2 - g_0^+) + 1] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+) + 1].
\end{aligned} \tag{25}$$

Let $\Delta_t(S, f)$ denote $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$. Without loss of generality, we assume that both P_1 and P_0 are positive. Otherwise, it can be verified that $\Delta_t(S, f) = 0$. Then, we get:

¹Note that the notations P_1 , P_0 , g_1^+ , and g_0^+ —as well as all others used throughout the paper—are not fixed constants, but rather depend on the specific input $\mathbf{x}_S$. Formally, they should be written as $P_1(\mathbf{x}_S)$, $P_0(\mathbf{x}_S)$, and so on. However, for readability, we omit the explicit dependence on $\mathbf{x}_S$.

$$\begin{aligned}
\Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot [2(P_1 \cdot (g_1^+)^2 + P_0 \cdot (g_0^+)^2) - 2(P_1 \cdot g_1^+ + P_0 \cdot g_0^+)^2] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot [(P_1 - P_1^2)(g_1^+)^2 + (P_0 - P_0^2)(g_0^+)^2 - 2 \cdot P_1 \cdot P_0 \cdot g_1^+ \cdot g_0^+] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 \cdot P_0 \cdot (g_1^+ - g_0^+)^2 \\
&\geq 0.
\end{aligned} \tag{26}$$

Note that $P_1 - P_1^2 = P_1(1 - P_1) = P_1 P_0$ and similarly: $P_0 - P_0^2 = P_0(1 - P_0) = P_0 P_1$. Using these identities along with the previous result, we conclude that $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, thereby concluding our proof. $\square$

Lemma 2. *The global sufficient value function v_{uff}^g is monotonic non-decreasing for functions with discrete multi-valued input domains and multiple output classes, under any data distribution.*

Proof. We next show that the monotonicity contribution is not an artifact of the Boolean setting. The same argument extends to finite input domains and finite output ranges; let $\mathcal{K}$ denote the set of output classes.

We can generalize equation 22 as follows:

$$\begin{aligned}
\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] &= \\
\sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right]
\end{aligned} \tag{27}$$

Given the condition $\mathbf{z}_S = \mathbf{x}_S$, suppose feature t can take r possible values $\{v_1, \dots, v_r\}$. Then, equation 23 can be generalized as follows:

$$\begin{aligned}
&\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\
&= \sum_{k=1}^r \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_t = v_k}} \Pr(\mathbf{x}_S) \cdot \Pr(z_t = v_k \mid \mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k))^2 \right] \right].
\end{aligned} \tag{28}$$

Let P_k denote $\Pr(z_t = v_k \mid \mathbf{x}_S)$, Let g_k^j denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_k)$, that is, z_t takes the value v_k and the output class is j . Then, we have

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \tag{29}$$

and

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) \right]$$

Combining these two implications, we can conclude that:

$$\begin{aligned}
1242 \quad & \Delta_t(S, f) \\
1243 \quad &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{k=1}^r P_k \cdot \left(\sum_{j \in \mathcal{K}} (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot g_k^j \right)^2 \right] \\
1244 \quad &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r P_k \cdot (g_k^j)^2 \right) - \sum_{j \in \mathcal{K}} \left(\sum_{k=1}^r (P_k \cdot g_k^j)^2 + \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right) \right] \quad (30) \\
1245 \quad &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \sum_{k \neq l} P_k \cdot P_l \cdot (g_k^j)^2 - \sum_{j \in \mathcal{K}} \sum_{k < l} 2 \cdot P_k \cdot P_l \cdot g_k^j \cdot g_l^j \right] \\
1246 \quad &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k \cdot P_l \cdot (g_k^j - g_l^j)^2 \right] \\
1247 \quad &\geq 0.
\end{aligned}$$

Hence, $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)]$ is monotone, which completes the proof for this setting.

□

Lemma 3. *The global sufficient value function v_{diff}^g is monotonic non-decreasing for functions with continuous input domains and multiple output classes, under any data distribution.*

Proof. We then extend monotonicity to continuous feature domains. One route is to partition each continuous domain into disjoint subdomains and adapt lemma 2; the mathematical contribution is unchanged, but the notation becomes integral rather than summation-based.

Alternatively, we provide a direct proof below. Assume all features lie in continuous domains, and let $p(\mathbf{x}_S, \mathbf{x}_{\bar{S}})$ denote the corresponding joint probability density function. Moreover, let $p(x_i)$ denote the marginal probability density function of x_i , and $\mathcal{X}_i$ denote the domain of feature i .

We define marginal density as

$$p(\mathbf{x}_S) := \int_{\mathbf{x}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} p(\mathbf{x}_S, \mathbf{x}_{\bar{S}}) d\mathbf{x}_{\bar{S}}, \quad (31)$$

where $\mathcal{X}_{\bar{S}}$ represents the domain of features in $\bar{S}$.

And the conditional density is defined as:

$$p(x_t \mid \mathbf{x}_S) := \frac{p(\mathbf{x}_S, x_t)}{p(\mathbf{x}_S)}. \quad (32)$$

The local probability of S given $\mathbf{x}$ is defined as

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \frac{\int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot p(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) d\mathbf{z}_{\bar{S}}}{p(\mathbf{x}_S)}, \quad (33)$$

This term can also be decomposed into:

$$\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S) = \int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \quad (34)$$

via an additional feature $t \notin S$.

We then modify equation 27 to

$$\begin{aligned}
1293 \quad & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
1294 \quad &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} (\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S))^2 \right] d\mathbf{x}_S, \quad (35) \\
1295 \quad &
\end{aligned}$$

while equation 28 is changed to

$$\begin{aligned} & \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p(z_t \mid \mathbf{x}_S) \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S. \end{aligned} \quad (36)$$

For simplicity, let $Y_j(t)$ denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = v_t)$, and $p_{t|S}$ denote $p(z_t \mid \mathbf{x}_S)$. Note that both $Y_j(t)$ and $p_{t|S}$ are associated with a fixed $\mathbf{x}_S$ ². We prove monotonicity as follows:

$$\begin{aligned} & \Delta_t(S, f) \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}})] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \left(\sum_{j \in \mathcal{K}} (\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_{S \cup \{t\}} = \mathbf{x}_{S \cup \{t\}}))^2 \right) dz_t \right] d\mathbf{x}_S - \\ & \quad \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{z_t \in \mathcal{X}_t} p_{t|S} \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_t = x_t) dz_t \right)^2 \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t)^2 dz_t - \left(\int_{\mathcal{X}_t} p_{t|S} \cdot Y_j(t) dz_t \right)^2 \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{2}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t)^2 - Y_j(t)Y_j(t')) dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_{t|S} \cdot p_{t'|S} \cdot (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (37)$$

□

Proposition 2. While the local probabilistic setting (for any δ) lacks monotonicity — i.e., the value functions v_{con}^ℓ and v_{suff}^ℓ are non-monotonic (Arenas et al., 2022; Izza et al., 2023; Subercaseaux et al., 2025; Izza et al., 2024b) — in the global probabilistic setting (also for any δ), both value functions v_{con}^g and v_{suff}^g are monotonic non-decreasing.

Proof. Together, lemmas 1 to 3 establish the full monotonicity result needed in the main paper: v_{suff}^g is monotone non-decreasing for any well-defined classification function, whether its features are Boolean, discrete, continuous, or mixed, and under any data distribution.

$$v_{\text{con}}^g(S) = 1 - v_{\text{suff}}^g(\bar{S}) \quad (38)$$

This implies that v_{con}^g is monotonic non-increasing with respect to adding features to $\bar{S}$. Equivalently, v_{con}^g is monotonic non-decreasing as features are removed from $\bar{S}$ and added to S , thus completing the proof.

□

E PROOF OF PROPOSITION 3

Proposition 3. While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{suff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{suff}^g is supermodular.

Proof. This proof establishes the supermodularity contribution for global sufficient explanations. As in the monotonicity proof, we start with Boolean functions and then generalize to multi-valued

²Changing $\mathbf{x}_S$ alters their values, we omit $\mathbf{x}_S$ in the notation for brevity.

discrete domains, continuous domains, and arbitrary well-defined classification functions. Unlike monotonicity, this property requires feature independence; we therefore also include a counterexample showing that supermodularity can fail without that assumption.

Lemma 4. *The global sufficient value function v_{uff}^g is supermodular for Boolean functions, under the assumption of feature independence.*

Proof. Particularly, we will prove that:

$$\Delta_t(S, f) \leq \Delta_t(S', f), \quad (39)$$

where $S \subseteq S'$ and $t \notin S'$. That is, $\Delta_t(S, f)$ is monotone.

Clearly, if $S' = S$, then $\Delta_t(S, f) \leq \Delta_t(S', f)$ holds. Let $S' = S \cup \{i\}$, where $i \notin S$ and $i \neq t$. For the local probability $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$, under the assumption of feature independence, we have that the following holds:

$$\begin{aligned} \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S) \\ = \Pr(z_i = 1) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1) + \\ \Pr(z_i = 1) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 1) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1) + \\ \Pr(z_i = 0) \cdot \Pr(z_t = 0) \cdot \Pr_{\mathbf{z}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0). \end{aligned} \quad (40)$$

Let P_1, P_0, Q_1 , and Q_0 denote $\Pr(z_t = 1), \Pr(z_t = 0), \Pr(z_i = 1)$, and $\Pr(z_i = 0)$, respectively. Moreover, let $g_{11}^+, g_{10}^+, g_{01}^+$, and g_{00}^+ denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 1), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 1, z_t = 0), \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 1)$, and $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid \mathbf{z}_S = \mathbf{x}_S, z_i = 0, z_t = 0)$, respectively.

We further decompose equation 26 into:

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [(Q_1 \cdot g_{11}^+ + Q_0 \cdot g_{01}^+) - (Q_1 \cdot g_{10}^+ + Q_0 \cdot g_{00}^+)]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+) + Q_0 \cdot (g_{01}^+ - g_{00}^+)]^2. \end{aligned} \quad (41)$$

Similarly, the following also holds:

$$\Delta_t(S \cup \{i\}, f) = \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot [Q_1 \cdot (g_{11}^+ - g_{10}^+)^2 + Q_0 \cdot (g_{01}^+ - g_{00}^+)^2]. \quad (42)$$

Thus, we get

$$\begin{aligned} \Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} 2 \cdot \Pr(\mathbf{x}_S) \cdot P_1 P_0 \cdot Q_1 Q_0 \cdot (g_{11}^+ - g_{10}^+ - g_{01}^+ + g_{00}^+)^2 \\ &\geq 0. \end{aligned} \quad (43)$$

This implies that $\Delta_t(S, f)$ is monotone, that is, $c(S \cup \{t\}, f) - c(S, f) \leq c(S' \cup \{t\}, f) - c(S', f)$. $\square$

Lemma 5. *The global sufficient value function v_{uff}^g is supermodular for functions with discrete multi-valued input domains and multiple output classes, under the assumption of feature independence.*

Proof. We extend the results of the previous section to multi-valued discrete functions, again with the assumption of feature independence. We use P_k to denote $\Pr(z_t = v_k)$, and $Q_{k'}$ to denote $\Pr(z_i = v_{k'})$. Moreover, we use $g_{k'k}^j$ to denote $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j \mid \mathbf{z}_S = \mathbf{x}_S, z_i = v_{k'}, z_t = v_k)$. We decompose equation 30 to

$$\begin{aligned} \Delta_t(S, f) &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} P_k P_l \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'k}^j - \sum_{k'=1}^{r'} Q_{k'} \cdot g_{k'l}^j \right]^2 \\ &= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \Pr(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} P_k P_l \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j) \right]^2. \end{aligned} \quad (44)$$

Likewise, we have

$$\begin{aligned}
\Delta_t(S \cup \{i\}, f) &= \sum_{k'=1}^{r'} \left[\sum_{\substack{\mathbf{x}_S \sim \mathcal{D}_S \\ x_i = v'_k}} \mathbf{Pr}(\mathbf{x}_S) \cdot \mathbf{Pr}(z_i = v'_k) \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \left[\sum_{k'=1}^{r'} Q_{k'} \cdot \sum_{j \in \mathcal{K}} \left(\sum_{k < l} P_k P_l \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right) \right] \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} P_k P_l \cdot \sum_{k'=1}^{r'} Q_{k'} \cdot (g_{k'k}^j - g_{k'l}^j)^2 \right].
\end{aligned} \tag{45}$$

Therefore, we get

$$\begin{aligned}
&\Delta_t(S \cup \{i\}, f) - \Delta_t(S, f) \\
&= \sum_{\mathbf{x}_S \sim \mathcal{D}_S} \mathbf{Pr}(\mathbf{x}_S) \cdot \sum_{j \in \mathcal{K}} \left[\sum_{k < l} \sum_{k' < l'} P_k P_l Q_{k'} Q_{l'} \cdot (g_{k'k}^j - g_{k'l}^j - g_{l'k}^j + g_{l'l}^j)^2 \right] \\
&\geq 0.
\end{aligned} \tag{46}$$

□

Proposition 3. *While the local probabilistic sufficient setting (for any δ) lacks supermodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{diff}^ℓ is not supermodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{diff}^g is supermodular.*

Proof. We can generalize the supermodularity results established in Lemma 5 to the continuous case, assuming that $\mathcal{D}_p$ satisfies feature independence. One approach is to partition the domain of the continuous features into two disjoint subdomains and then adapt the proofs in Lemma 5 by revising the notation accordingly.

Alternatively, we provide a direct proof below. Under this assumption, the marginal and conditional densities simplify as follows:

$$p(\mathbf{x}_S) = \prod_{i \in S} p(x_i), \text{ and } p(x_t | \mathbf{x}_S) = p(x_t). \tag{47}$$

Consequently, the local probability of S conditioned on $\mathbf{x}$ becomes:

$$\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S) = \int_{\mathbf{z}_{\bar{S}} \in \mathcal{X}_{\bar{S}}} I(f(\mathbf{x}_S, \mathbf{z}_{\bar{S}}) = j) \cdot \prod_{i \in \bar{S}} p(z_i) d\mathbf{z}_{\bar{S}}. \tag{48}$$

We let $Y_j(i, t)$ denote $\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = j | \mathbf{z}_S = \mathbf{x}_S, z_i = x_i, z_t = x_t)$, where $i \notin S, t \notin S$ and $i \neq t$. Evidently,

$$Y_j(i, t) = Y_j(t, i), \text{ and } Y_j(t) = \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i.$$

We have:

$$\begin{aligned}
&\Delta_t(S, f) \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} (Y_j(t) - Y_j(t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot Y_j(i, t) dz_i - \int_{\mathcal{X}_i} p_i \cdot Y_j(i, t') dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\
&= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_S,
\end{aligned} \tag{49}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \left(\int_{\mathcal{X}_i} p_i \cdot (Y_j(i, t) - Y_j(i, t')) dz_i \right)^2 dz_t dz_{t'} \\ &= \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} \cdot (Y_j(i, t) - Y_j(i, t')) \cdot (Y_j(i', t) - Y_j(i', t')) dz_i dz_{i'} dz_t dz_{t'}. \end{aligned}$$

Let $S' = S \cup \{i\}$, then we have:

$$\begin{aligned} & \Delta_t(S', f) \\ &= \int_{\mathbf{x}_{S'} \in \mathcal{X}_{S'}} p(\mathbf{x}_{S'}) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) \right] d\mathbf{x}_{S'} \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \int_{\mathcal{X}_i} p_i \cdot \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} \right) dz_i \right] d\mathbf{x}_S \quad (50) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \cdot \left[\sum_{j \in \mathcal{K}} \frac{1}{2} \left(\int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \right) \right] d\mathbf{x}_S, \end{aligned}$$

where

$$\begin{aligned} & \int_{\mathcal{X}_i} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i \\ &= \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} \int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} p_i p_{i'} p_t p_{t'} \cdot (Y_j(i, t) - Y_j(i, t'))^2 dz_t dz_{t'} dz_i dz_{i'}. \end{aligned}$$

Let $\alpha(i, t, t')$ denote $Y_j(i, t) - Y_j(i, t')$ and $\beta(i', t, t')$ denote $Y_j(i', t) - Y_j(i', t')$, we can infer that

$$\begin{aligned} & \Delta_t(S', f) - \Delta_t(S, f) \\ &= \int_{\mathbf{x}_S \in \mathcal{X}_S} p(\mathbf{x}_S) \left[\sum_{j \in \mathcal{K}} \frac{1}{4} \left(\int_{\mathcal{X}_t} \int_{\mathcal{X}_{t'}} \int_{\mathcal{X}_i} \int_{\mathcal{X}_{i'}} p_t p_{t'} p_i p_{i'} (\alpha(i, t, t') - \beta(i', t, t'))^2 dz_i dz_{i'} dz_t dz_{t'} \right) \right] d\mathbf{x}_S \\ &\geq 0. \end{aligned} \quad (51)$$

Consequently, the global sufficient value function v_{suff}^g retains the supermodularity property for any valid classification function, regardless of whether its features are Boolean, discrete, continuous, or a mix thereof.

We now demonstrate that the local probabilistic sufficient value function is neither submodular nor supermodular. As a simple illustration, fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define a Boolean function as follows:

$$f(x_1, x_2) = (x_1 \vee x_2) \wedge x_3. \quad (52)$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic sufficient value function v_{suff}^l , we have that the following holds:

$$\begin{aligned} v_{\text{suff}}^l(\emptyset) &= \frac{3}{8}, \quad v_{\text{suff}}^l(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{2\}) = \frac{1}{2}, \quad v_{\text{suff}}^l(\{3\}) = \frac{3}{4}, \\ v_{\text{suff}}^l(\{1, 2\}) &= \frac{1}{2}, \quad v_{\text{suff}}^l(\{1, 3\}) = 1, \quad v_{\text{suff}}^l(\{2, 3\}) = 1, \quad v_{\text{suff}}^l(\{1, 2, 3\}) = 1. \end{aligned}$$

Since $(v_{\text{suff}}^l(\{1, 2\}) - v_{\text{suff}}^l(\{2\})) < (v_{\text{suff}}^l(\{1\}) - v_{\text{suff}}^l(\emptyset)) < (v_{\text{suff}}^l(\{1, 3\}) - v_{\text{suff}}^l(\{3\}))$, the function v_{suff}^l is neither supermodular nor submodular. □

F PROOF OF PROPOSITION 4

Proposition 4. *While the local probabilistic contrastive setting (for any δ) lacks submodularity — even when $\mathcal{D}$ is uniform, i.e., the value function v_{con}^l is not submodular — in the global probabilistic setting (also for any δ), when $\mathcal{D}$ exhibits feature independence, the value function v_{con}^g is submodular.*

Proof. The submodularity contribution for global contrastive explanations follows by dualizing the supermodularity result for global sufficiency. Let $S \subseteq [n]$ be the set of features allowed to change, and suppose $S \subseteq S'$ with $i \notin S'$. To compare the marginal contributions $v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)$ and $v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')$, we equivalently compare $(1 - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}))$ and $(1 - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) - (1 - v_{\text{suff}}^g(\bar{S}'))$. Since $S \subseteq S'$, it follows that $\bar{S}' \subseteq \bar{S}$, and therefore we obtain:

$$(v_{\text{suff}}^g(\bar{S}) - v_{\text{suff}}^g(\bar{S} \setminus \{i\})) \geq (v_{\text{suff}}^g(\bar{S}') - v_{\text{suff}}^g(\bar{S}' \setminus \{i\})) \quad (53)$$

Which is equivalent to:

$$(v_{\text{con}}^g(S \cup \{i\}) - v_{\text{con}}^g(S)) \geq (v_{\text{con}}^g(S' \cup \{i\}) - v_{\text{con}}^g(S')). \quad (54)$$

Thus, the global contrastive value function v_{con}^g is submodular, hence concluding this proof.

To complete the local/global contrast, we show that the local probabilistic contrastive value function has neither submodularity nor supermodularity. Fix the uniform distribution on the three variables $\{x_1, x_2, x_3\}$ and define the Boolean function:

$$f(x_1, x_2) := (x_1 \vee x_2) \wedge x_3. \quad (55)$$

Consider the input $\mathbf{x} = (1, 1, 1)$ which is classified as 1. For the local probabilistic contrastive value function v_{con}^l , we have:

$$\begin{aligned} v_{\text{con}}^l(\emptyset) &= 1, \quad v_{\text{con}}^l(\{1\}) = 1, \quad v_{\text{con}}^l(\{2\}) = 1, \quad v_{\text{con}}^l(\{3\}) = \frac{1}{2}, \\ v_{\text{con}}^l(\{1, 2\}) &= \frac{3}{4}, \quad v_{\text{con}}^l(\{1, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{2, 3\}) = \frac{1}{2}, \quad v_{\text{con}}^l(\{1, 2, 3\}) = \frac{3}{8}. \end{aligned}$$

Since $(v_{\text{con}}^l(\{1, 2\}) - v_{\text{con}}^l(\{2\})) < (v_{\text{con}}^l(\{1, 2, 3\}) - v_{\text{con}}^l(\{2, 3\})) < (v_{\text{con}}^l(\{1\}) - v_{\text{con}}^l(\emptyset))$, the function v_{con}^l is neither submodular nor supermodular. □

In the rest of this section, we show that the feature independence assumption is not just sufficient but also necessary for supermodularity. We illustrate this with the following counterexample.

Define a Boolean function:

$$f(x_1, x_2) := x_1 \quad (56)$$

For this boolean function, it holds that $f(00) = 0$, $f(01) = 0$, $f(10) = 1$, and $f(11) = 1$. Moreover, define a distribution $\mathcal{D}$ on x_1 and x_2 . Recall that the global sufficient value function is:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] = \sum_{\mathbf{x} \sim \mathcal{D}} \Pr(\mathbf{x}) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)$$

In addition, if we can show that

$$\Delta_t(S, f) \leq \Delta_t(S', f),$$

where $S \subseteq S'$ and $t \notin S'$, then the global sufficient value function is supermodular. When $S = \emptyset$, we have that the following holds:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\ = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0) + \\ \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) + \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1) \end{aligned}$$

Which is also equivalent to:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)) \\
& = (\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1))^2 + (\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1))^2
\end{aligned}$$

When $S = \{2\}$, we have that the following condition holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_2 = 1) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 0) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_2 = 1)
\end{aligned}$$

This can also be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)} \\
& = \frac{\Pr(x_1 = 0, x_2 = 0)^2 + \Pr(x_1 = 1, x_2 = 0)^2}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 1, x_2 = 0)} + \frac{\Pr(x_1 = 0, x_2 = 1)^2 + \Pr(x_1 = 1, x_2 = 1)^2}{\Pr(x_1 = 0, x_2 = 1) + \Pr(x_1 = 1, x_2 = 1)}
\end{aligned}$$

In contrast, when $S = \{1\}$, we have the the following holds:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
& = \Pr(x_1 = 0, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0) + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1) + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1)
\end{aligned}$$

This can similarly be rearranged as follows:

$$\begin{aligned}
& \Pr(x_1 = 0, x_2 = 0) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 0, x_2 = 1) \cdot \frac{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)}{\Pr(x_1 = 0, x_2 = 0) + \Pr(x_1 = 0, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 0) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} + \\
& \Pr(x_1 = 1, x_2 = 1) \cdot \frac{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)}{\Pr(x_1 = 1, x_2 = 0) + \Pr(x_1 = 1, x_2 = 1)} = 1
\end{aligned}$$

Finally, when $S = \{1, 2\}$, we have the following condition:

$$\begin{aligned}
& \mathbb{E}_{\mathbf{z} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = f(\mathbf{x}) \mid \mathbf{z}_S = \mathbf{x}_S)] \\
&= \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 0 \mid x_1 = 0, x_2 = 1) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 0) + \\
&\quad \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) = 1 \mid x_1 = 1, x_2 = 1)
\end{aligned}$$

This can likewise be reordered as follows:

$$\begin{aligned}
& \mathbf{Pr}(x_1 = 0, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)}{\mathbf{Pr}(x_1 = 0, x_2 = 0)} + \mathbf{Pr}(x_1 = 0, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)}{\mathbf{Pr}(x_1 = 0, x_2 = 1)} + \\
& \mathbf{Pr}(x_1 = 1, x_2 = 0) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 0)}{\mathbf{Pr}(x_1 = 1, x_2 = 0)} + \mathbf{Pr}(x_1 = 1, x_2 = 1) \cdot \frac{\mathbf{Pr}(x_1 = 1, x_2 = 1)}{\mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= 1
\end{aligned}$$

Let $t = 2$. Evidently, we have $\Delta_t(\{1\}, f) = 0$. Next, let us analyse $\Delta_t(\emptyset, f)$. We note that if we assume feature independence, we observe that:

$$\begin{aligned}
& (\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 0, x_2 = 1))^2 + (\mathbf{Pr}(x_1 = 1, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 1))^2 \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2,
\end{aligned}$$

Additionally, another condition that holds is:

$$\begin{aligned}
& \frac{\mathbf{Pr}(x_1 = 0, x_2 = 0)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 0)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 0) + \mathbf{Pr}(x_1 = 1, x_2 = 0)} + \frac{\mathbf{Pr}(x_1 = 0, x_2 = 1)^2 + \mathbf{Pr}(x_1 = 1, x_2 = 1)^2}{\mathbf{Pr}(x_1 = 0, x_2 = 1) + \mathbf{Pr}(x_1 = 1, x_2 = 1)} \\
&= \mathbf{Pr}(x_1 = 0)^2 + \mathbf{Pr}(x_1 = 1)^2.
\end{aligned}$$

This implies $\Delta_t(\emptyset, f) = 0$, that is, $\Delta_t(\emptyset, f) \leq \Delta_t(\{1\}, f)$. However, without the assumption of feature independence, we cannot guarantee $\Delta_t(\emptyset, f) = 0$. For example, let:

$$\mathbf{Pr}(x_1 = 0, x_2 = 0) = P_{00}, \mathbf{Pr}(x_1 = 0, x_2 = 1) = P_{01}$$

and

$$\mathbf{Pr}(x_1 = 1, x_2 = 0) = P_{10}, \mathbf{Pr}(x_1 = 1, x_2 = 1) = P_{11}$$

Then, by simplifying $\Delta_t(\emptyset, f)$ we get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{P_{00}^2 + P_{10}^2}{P_{00} + P_{10}} + \frac{P_{01}^2 + P_{11}^2}{P_{01} + P_{11}} \right] - \left[(P_{00} + P_{01})^2 + (P_{10} + P_{11})^2 \right] \quad (57)$$

We then let $P_{00} = 0.4$, $P_{01} = 0.2$, $P_{10} = 0.1$, and $P_{11} = 0.3$, and get that the following holds:

$$\Delta_t(\emptyset, f) = \left[\frac{(0.4)^2 + (0.1)^2}{0.5} + \frac{(0.2)^2 + (0.3)^2}{0.5} \right] - \left[(0.4 + 0.2)^2 + (0.1 + 0.3)^2 \right] = 0.08. \quad (58)$$

This implies that $\Delta_t(\emptyset, f) > \Delta_t(1, f)$, indicating that — unlike in the monotonicity setting where it was not required — feature independence is a necessary condition for supermodularity.

G PROOF OF PROPOSITION 5

Proposition 5. *Computing Algorithm 1 with the local value functions v_{con}^ℓ and v_{suff}^ℓ does not always converge to a subset minimal δ -sufficient/contrastive reason. However, computing it with the global value functions v_{con}^g or v_{suff}^g necessarily produces subset minimal δ -sufficient/contrastive reasons.*

Proof. This proposition is the algorithmic payoff of the monotonicity contribution. Since Proposition 2 proves that both v_{suff}^g and v_{con}^g are monotone non-decreasing, the greedy deletion algorithm has the structure needed to certify subset minimality. The rest of the proof explains this implication and contrasts it with the local case.

For v_{con}^g and v_{suff}^g , Algorithm 1 converges to a subset-minimal global δ sufficient or contrastive reason. Prior work (Ignatiev et al., 2019; Wu et al., 2024b; Arenas et al., 2022) shows that this greedy deletion strategy is sound for monotone non-decreasing value functions. The algorithm stops only when every single-feature deletion $v(S \setminus \{i\})$ falls below δ , while the invariant $v(S) \geq \delta$ is maintained. Monotonicity then rules out every proper subset of S , so the returned explanation is subset-minimal.

The local case lacks precisely this certificate. The local probabilistic sufficient value function, and likewise the local probabilistic contrastive value function, is not monotone Arenas et al. (2022); Izza et al. (2023); Subercaseaux et al. (2025); Izza et al. (2024b). As a simple illustration, fix the uniform distribution on the two binary variables $\{x_1, x_2\}$ and define:

$$f(x_1, x_2) := x_1 \vee x_2. \quad (59)$$

Consider the input $\mathbf{x} = (0, 1)$ which is classified as 1, then:

$$v_{\text{suff}}^{\ell}(\emptyset) = \frac{3}{4}, \quad v_{\text{suff}}^{\ell}(\{1\}) = \frac{1}{2}, \quad v_{\text{suff}}^{\ell}(\{2\}) = 1, \quad v_{\text{suff}}^{\ell}(\{1, 2\}) = 1. \quad (60)$$

Because $v_{\text{suff}}^{\ell}(\emptyset) > v_{\text{suff}}^{\ell}(\{1\}) < v_{\text{suff}}^{\ell}(\{1, 2\})$, the function v_{suff}^{ℓ} is not monotone. Likewise, it can be computed that:

$$v_{\text{con}}^{\ell}(\emptyset) = 1, \quad v_{\text{con}}^{\ell}(\{1\}) = 1, \quad v_{\text{con}}^{\ell}(\{2\}) = \frac{1}{2}, \quad v_{\text{con}}^{\ell}(\{1, 2\}) = \frac{3}{4}. \quad (61)$$

(For v_{con}^{ℓ} , the S denotes the set of features that are allowed to vary.) Because $v_{\text{con}}^{\ell}(\emptyset) > v_{\text{con}}^{\ell}(\{2\}) < v_{\text{con}}^{\ell}(\{1, 2\})$, the function v_{con}^{ℓ} is also non-monotone.

□

H PROOF OF THEOREM 1

Theorem 1. *If f is a decision tree and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time. However, unless $\text{PTIME}=\text{NP}$, no polynomial-time algorithm exists for computing a local δ -sufficient reason for decision trees even under independent distributions.*

Proof. The decision-tree result combines two ingredients from the main contribution structure: efficient computation of the global value term and monotonicity of that term. Under independent distributions, the relevant decision-tree value function is:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (62)$$

This value can be computed in polynomial time. The tree structure lets us iterate over all pairs of leaves, each representing partial assignments $\mathbf{x}_S$ and $\mathbf{z}_{S'}$. For any pair with the same prediction under f , we compute its contribution to $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\mathbf{Pr}_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ by multiplying the corresponding feature-wise probabilities over shared features. Feature independence makes these probabilities factor, and summing over all matching leaf pairs gives the full expectation.

All probabilities are part of the input and every operation above is polynomial, so each call made by Algorithm 1 is efficient. Lemma 2 then supplies the correctness guarantee: v_{suff}^g is monotone non-decreasing, so once the algorithm halts with every deletion $v(S \setminus \{i\})$ below δ and $v(S) \geq \delta$ preserved, no strict subset can satisfy the threshold. Thus the output is subset-minimal and the full procedure is polynomial time.

For the remaining part of the claim, the result follows from (Arenas et al., 2022), which showed that, assuming $P \neq NP$, there is no polynomial-time algorithm for computing a subset-minimal δ -sufficient reason for decision trees. While this was proven specifically under the uniform distribution, the hardness clearly extends to independent distributions, which include this case. Combined, these results establish the complexity separation between the local and global variants, thus concluding our proof.

□

I PROOF OF PROPOSITION 6

Proposition 6. *For any model f , and empirical distribution $\mathcal{D}$ — computing a subset-minimal global δ -sufficient or δ -contrastive reason for f can be done in polynomial time.*

Proof. This proof explains why the global subset-minimal tractability result extends to empirical distributions and arbitrary polynomial-time inference models. The computation of the global sufficient probability function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (63)$$

or the computation of the global contrastive probability function

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (64)$$

can be performed in polynomial time when $\mathcal{D}$ is empirical. We iterate over pairs of instances $\mathbf{x}, \mathbf{z}$ in the dataset and evaluate $f(\mathbf{x})$ and $f(\mathbf{z})$ to compute the relevant expectation. Hence, threshold testing against δ is also polynomial time. Combining this value-oracle computation with the global monotonicity result (Proposition D) lets Algorithm 1 search for either sufficient or contrastive reasons using only a linear number of polynomial-time queries. The resulting subset-minimal global explanation is therefore computable in polynomial time.

□

J PROOF OF THEOREM 2

Theorem 2. *Assuming f is a neural network or a tree ensemble, and $\mathcal{D}$ is an empirical distribution — there exist polynomial-time algorithms for obtaining subset minimal global δ -sufficient and contrastive reasons. However, unless $PTIME=NP$, there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason or a subset minimal local δ -contrastive reason.*

Proof. This proof packages the empirical-distribution local/global separation. The global tractability side follows from Proposition 6: for any model with polynomial-time inference, both $v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ and $v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)]$ can be computed in polynomial time from the empirical distribution. For the local hardness side, we prove two lemmas: one for sufficient reasons and one for contrastive reasons.

Lemma 6. *Unless $PTIME=NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -sufficient reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will first prove this claim for neural networks and then extend the result to tree ensembles. We will establish this claim by demonstrating that if a polynomial-time algorithm exists for computing a subset-minimal local δ sufficient reason for neural networks, then it would enable us to solve the classic NP-complete *CNF-SAT* problem in polynomial time. The *CNF-SAT* problem is defined as follows:

CNF-SAT:

Input: A formula in conjunctive normal form (CNF): ϕ .

Output: Yes, if there exists an assignment to the n literals of ϕ such that ϕ is evaluated to True, and No otherwise

Our proof will also utilize the following lemma (with its proof provided in Barceló et al. (2020)):

Lemma 7. *Any boolean circuit ϕ can be encoded into an equivalent MLP over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We will actually establish hardness for a simpler, more specific case, which will consequently imply hardness for the more general setting. In this case, we assume that the empirical distribution $\mathcal{D}$ consists of a single element, which, for simplicity, we take to be the zero vector $\mathbf{0}_n$. Since the sufficiency of S in this scenario depends only on the local instance $\mathbf{x}$ and the single instance $\mathbf{0}_n$ in our empirical distribution, the nature of the input format (whether discrete, continuous, etc.) does not affect the result. Therefore, the hardness results hold universally across all these settings.

Similarly to $\mathbf{0}_n$, we define $\mathbf{1}_n$ as an n -dimensional vector consisting entirely of ones. Given an input ϕ , we initially assign $\mathbf{1}_n$ to ϕ , effectively setting all variables to True. If ϕ evaluates to True under this assignment, then a satisfying truth assignment exists. Therefore, we assume that $\phi(\mathbf{1}_n) = 0$. We now introduce the following definitions:

$$\begin{aligned}\phi_2 &:= (x_1 \wedge x_2 \wedge \dots \wedge x_n), \\ \phi' &:= \phi \vee \phi_2\end{aligned}\tag{65}$$

ϕ' , while no longer a CNF, can still be transformed into an MLP f using Lemma 8, ensuring that f behaves equivalently to ϕ' over the domain $\{0, 1\}^n$. Given our assumption that computing a subset-minimal local δ -sufficient reason is feasible in polynomial time, we can determine one for the instance $\langle f, \mathbf{x} := \mathbf{1}_n \rangle$, $\delta := 1$, noting that the empirical distribution $\mathcal{D}$ is simply defined over the single data point $\mathbf{0}_n$.

We now assert that the subset-minimal δ -sufficient reason generated for $\langle f, \mathbf{x} \rangle$ encompasses the entire input space, i.e., $S = \{1, \dots, n\}$, if and only if $\langle \phi \rangle \notin \text{CNF-SAT}$.

Let us assume that $S = \{1, \dots, n\}$. Since S is a δ -sufficient reason for $\langle f, \mathbf{x} \rangle$, this simply means that setting the complementary set to any value maintains the prediction. Since the complementary set is $\emptyset$ in this case, this trivially holds. The fact that S is subset-minimal means that any other subset $S' \subseteq S$ satisfies $v(S') < \delta = 1$. Since the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$ is determined by a single point (the distribution contains only the point $\mathbf{0}_n$), the probability function can only take values of 1 or 0. Hence, we also know that $v(S') = 0$. This tells us that, aside from the subset $S = \{1, \dots, n\}$, for any subset $S' \subseteq S$, fixing the features in S' to 1 and the rest to 0 does not result in a classification outcome of 1. Since the ϕ_2 component within ϕ' is True only if all features are assigned 1, this directly implies that ϕ is assigned False for any of these inputs. Since we already know that ϕ does not return a True answer for the vector assignment $\mathbf{1}_n$ (as verified at the beginning), and now we have established that the same holds for all other input vectors, we conclude that $\phi \notin \text{CNF-SAT}$.

Now, suppose that S is a subset that is strictly contained within $\{1, \dots, n\}$. Given that S is sufficient under our definitions of the distribution $\mathcal{D}$ with $\delta = 1$, we can apply the same reasoning as before to conclude that $v(S) = 1$. This implies that setting the features in S to 1 while setting the remaining features to 0 ensures that the function f evaluates to 1. Since this assignment is necessarily not a vector consisting entirely of ones, it follows that the ϕ_2 component within ϕ' must be False. Consequently, the ϕ component must be True, which implies that $\langle \phi \rangle \in \text{CNF-SAT}$. This completes the proof.

We have established that the following claim holds for neural networks. However, extending the proof to tree ensembles requires a minor and straightforward adaptation. To do so, we will utilize the following lemma, which has been noted in several previous works (Ordyniak et al., 2024; Audemard et al., 2022b):

Lemma 8. *Any CNF or DNF ϕ can be encoded into an equivalent random forest classifier over the binary domain $\{0, 1\}^n \rightarrow \{0, 1\}$ in polynomial time.*

Proof. We observe that we can apply the same process used in our proof for neural networks, where we encoded ϕ' into an equivalent neural network. However, ϕ' is no longer a valid CNF due to our construction (though encoding it into an MLP was not an issue, as any Boolean circuit can be transformed into an MLP). Nevertheless, since ϕ' consists of a conjunction of only two terms, we can easily represent it as an equivalent CNF:

$$\phi' := (c_1 \vee \phi_2) \wedge (c_2 \vee \phi_2) \wedge \dots (c_m \vee \phi_2) \quad (66)$$

Where each c_i is a disjunction of a few terms. Consequently, ϕ' is a valid CNF, allowing us to transform it into an equivalent random forest classifier. The reduction we outlined for MLPs applies directly to these models as well, thereby completing the proof for both model families.

□

Lemma 9. *Unless $PTIME=NP$, then there is no polynomial time algorithm for computing a subset minimal local δ -contrastive reason for either a tree ensemble or a neural network, under empirical distributions.*

Proof. We will present a proof analogous to the one in Lemma 6. Specifically, we will once again utilize the classical NP-hard CNF-SAT problem defined in Lemma 6. In particular, given a Boolean formula ϕ , we will demonstrate that determining a subset-minimal contrastive reason — whether for a neural network or a tree ensemble — allows us to decide the satisfiability of ϕ .

First, we check whether assigning all variables in ϕ to 1 evaluates ϕ to True. If so, the formula is satisfiable, and we have determined its satisfiability. Otherwise, we use Lemma 8 to encode the CNF formula as a neural network f . Next, we compute a subset-minimal δ -sufficient reason by setting $\mathcal{D}$ as the empirical distribution containing only a single data point $\mathbf{0}_n$, following a similar procedure to Lemma 6. Additionally, we set $\delta = 1$ and compute a subset-minimal δ -contrastive reason concerning $\langle f, \mathbf{x} \rangle$.

We will now demonstrate that if any subset-minimal δ contrastive reason obtained for ϕ is satisfiable, then ϕ itself is satisfiable. Conversely, if no subset-minimal δ contrastive reason is obtained, then ϕ is unsatisfiable. The validity of this claim follows a reasoning similar to that provided in Lemma 6. Specifically, the term $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)$, where the distribution $\mathcal{D}$ considers sampling from a single datapoint, can set the probability to either 0 or 1. Furthermore, since we are searching for a $\delta = 1$ contrastive reason, this is equivalent to asking whether there exists an assignment that changes the classification of f , which corresponds to modifying the assignment of ϕ from False to True. If such an assignment exists, then ϕ is satisfiable. However, if no subset-minimal contrastive reason exists, then no subset of features fixed to zero — when the complementary set is set to ones — evaluates to true. This is equivalent to stating that no assignment evaluates f (and consequently ϕ) to True, implying that ϕ is unsatisfiable.

To extend the proof from neural networks to tree ensembles, we can follow the same procedure outlined in Lemma 6, encoding the CNF formula into an equivalent random forest classifier. Consequently, the proof remains valid for tree ensembles, thereby concluding the proof.

□

K PROOF OF THEOREM 3

Theorem 3. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence. In contrast, unless $PTIME=NP$, no bounded approximation exists for computing a local cardinally minimal δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The proof is split to match the theorem’s contribution structure: one lemma proves the global approximation guarantee, and the other proves the absence of any bounded local approximation.

Lemma 10. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O}\left(\ln\left(\frac{v_{con}^g([n])}{\min_{i \in [n]} v_{con}^g(\{i\})}\right)\right)$ -approximation, bounded by $\mathcal{O}(\ln(|D|))$, for computing a global cardinally minimal δ -contrastive reason for f , assuming feature independence.*

Proof. We prove a stronger model-agnostic claim. The argument only requires polynomial-time inference and one additional preprocessing condition introduced below; both requirements are satisfied by the neural-network and tree-ensemble formalizations.

We begin by noting that, since we are working with empirical distributions, the computation of the global contrastive probability value function:

$$v_{\text{con}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})] \quad (67)$$

Can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset $\mathcal{D}$, as previously established in Proposition 2. Since Algorithm 2 performs only a linear number of these polynomial-time queries, its total runtime is therefore polynomial.

Regarding the approximation, the classical work by Wolsey et al. (Wolsey, 1982) established a harmonic-series-based approximation guarantee for monotone non-decreasing submodular functions v with integer values. More generally, their result yields an approximation factor of $\mathcal{O}\left(\ln\left(\frac{v([n])}{\min_{i \in [n]} v(\{i\})}\right)\right)$. We showed in Proposition 2 that the value function v_{con}^g is monotone non-decreasing, and under feature independence, Proposition 4 establishes its submodularity. Combined with the fact that Algorithm 2 runs in polynomial time, this directly yields an approximation guarantee of $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$.

However, we must also show that the expression $\mathcal{O}\left(\ln\left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})}\right)\right)$ is both finite and bounded by $\mathcal{O}(\ln(|\mathcal{D}|))$, implying that it is effectively constant, due to the assumption of a fixed dataset $|\mathcal{D}|$. To ensure this, we begin by confirming that the expression is well-defined and finite. We then proceed to establish the desired bounds. To do so, we introduce a preprocessing step in which we eliminate “redundant” elements—those that could theoretically cause the denominator $\min_{i \in [n]} v_{\text{con}}^g(\{i\})$ to be zero. We begin by formally defining what we mean by redundancy:

Definition 1. Let $\mathcal{D}$ denote some empirical distribution over a dataset $\mathcal{D}$. Then we say that some feature $i \in [n]$ is redundant with respect to $\mathcal{D}$ if for any pair $\mathbf{x}, \mathbf{z} \in \mathcal{D}$ it holds that $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) = f(\mathbf{x})$.

Here, the notation $f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}})$ indicates that all features in $[n] \setminus \{i\}$ are fixed to their values in $\mathbf{x}$, while feature i is set to its value in $\mathbf{z}$. Notably, this is equivalent to defining:

$$\forall S \subseteq [n], \mathbf{z} \in \mathcal{D} \quad f(\mathbf{x}_S; \mathbf{z}_{\bar{S}}) = f(\mathbf{x}_{S \setminus \{i\}}; \mathbf{z}_{\bar{S} \cup \{i\}}) \quad (68)$$

As before, the notation $(\mathbf{x}_S; \mathbf{z}_{\bar{S}})$ indicates that the features in S are fixed to their values in $\mathbf{x}$, while the features in $\bar{S}$ are fixed to their values in $\mathbf{z}$. Thus, if we “remove” feature i from the input space and define a new function f' over the reduced space $[n'] := [n] \setminus \{i\}$, then every input of size n' receives the same output from f' as f gives on the corresponding remaining features $[n] \setminus \{i\}$. This removal can be done in polynomial time for both tree ensembles and neural networks: for neural networks, it detaches the corresponding input neuron; for tree ensembles, it removes splits on that feature from all trees. Because $v_{\text{con}}^g(S)$ is empirical, as discussed above and in Proposition 2, we can compute each $v_{\text{con}}^g(\{i\})$ by iterating over all pairs $\mathbf{x}, \mathbf{z} \in \mathcal{D}$ and checking whether:

$$f(\mathbf{x}_{[n] \setminus \{i\}}; \mathbf{z}_{\{i\}}) \neq f(\mathbf{x}) \quad (69)$$

If this condition holds, it indicates that feature i is not redundant. Conversely, if the condition fails for all pairs considered during iteration, then i is deemed redundant. Once all redundant features have been identified with respect to the empirical distribution $\mathcal{D}$, we can remove them and construct an equivalent model f' . As discussed above, this transformation can be performed in polynomial time for both neural networks and decision trees.

To conclude our proof, we observe that $v_{\text{con}}^g(\{i\})$ equals 0 for some empirical distribution $\mathcal{D}$ if and only if feature i is redundant with respect to $\mathcal{D}$. Therefore, once the preprocessing step removes all redundant features from f and we construct the resulting function f' , we ensure that $\min_{i \in [n]} v_{\text{con}}^g(\{i\}) > 0$.

We now present the more precise bounds referenced in our proof. Specifically, we demonstrate that the approximation factor is $\mathcal{O}(\ln(|\mathcal{D}|))$. Given that the empirical distribution $\mathcal{D}$ is fixed, this yields a constant approximation. While tighter bounds may be achievable, our goal here is solely to establish

that the bound is *constant* — a property that will later sharply contrast with the local explainability setting, where no bounded approximation exists.

More specifically, we know that the probability function $\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})$ is, by the definition of empirical distributions, at least $\frac{1}{|\mathcal{D}|}$. Therefore, the definition of $\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) \neq f(\mathbf{z}) \mid \mathbf{x}_{\bar{S}} = \mathbf{z}_{\bar{S}})]$ (i.e., the value function v_{con}^g) is at least $\frac{1}{|\mathcal{D}|^2}$. In particular, this also implies that:

$$\min_{i \in [n]} v_{\text{con}}^g(\{i\}) \geq \frac{1}{|\mathcal{D}|^2} \quad (70)$$

Consequently, we obtain that:

$$\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right) \leq \ln(|\mathcal{D}|^2) = \mathcal{O}(\ln(|\mathcal{D}|)) \quad (71)$$

Which concludes our proof.

□

Lemma 11. *Unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a cardinally minimal local δ -contrastive reason for any $\langle f, \mathbf{x} \rangle$ where f is either a neural network or a tree ensemble, even when $|D| = 1$.*

Proof. Assume, for contradiction, that there exists a bounded approximation algorithm for computing a cardinally minimal local δ -contrastive reason for some $\langle f, \mathbf{x} \rangle$, where f is a neural network or a tree ensemble—even when $|D| = 1$. However, Lemma 9 establishes that even with a single baseline $\mathbf{z} = 0_n$ (i.e., when the entire dataset is just one instance), deciding whether a contrastive reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Therefore, if a polynomial-time algorithm could yield a bounded approximation for a cardinally minimal contrastive reason, it would contradict this result, as such an approximation would implicitly decide the existence of a contrastive explanation.

□

L PROOF OF THEOREM 4

Theorem 4. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence. In contrast, unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|D| = 1$.*

Proof. The proof is split to mirror the theorem’s contribution structure: one lemma proves the global approximation guarantee, and the other proves the local inapproximability result.

Lemma 12. *Given a neural network or tree ensemble f and an empirical distribution $\mathcal{D}$ over a fixed dataset D , Algorithm 2 yields a constant $\mathcal{O} \left(\frac{1}{1-k^f} + \ln \left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})} \right) \right)$ -approximation for computing a global cardinally minimal δ -sufficient reason for f , assuming feature independence.*

Proof. The sufficient-reason approximation proof parallels Lemma 10, but uses the supermodular guarantee rather than the submodular one. For contrastive reasons we obtained the factor $\ln \left(\frac{v_{\text{con}}^g([n])}{\min_{i \in [n]} v_{\text{con}}^g(\{i\})} \right)$, which becomes finite and bounded by $\mathcal{O}(\ln(|\mathcal{D}|))$ after preprocessing. Here we show the analogous bounded guarantee for v_{suff}^g .

As in Lemma 10, we prove a stronger model-agnostic statement. The argument only requires polynomial-time inference and the redundant-feature removal condition described in Lemma 10; both are satisfied by the neural-network and tree-ensemble formalizations used in the theorem.

Here as well, since we are working with empirical distributions, the computation of the global sufficient probability value function:

$$v_{\text{suff}}^g(S) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{x}) = f(\mathbf{z}) \mid \mathbf{x}_S = \mathbf{z}_S)] \quad (72)$$

can be computed in polynomial time by iterating over all pairs $\mathbf{x}, \mathbf{z}$ in the dataset $\mathcal{D}$, as previously shown in both Lemma 10 and Proposition 2. Since Algorithm 2 makes only a linear number of such polynomial-time queries, its overall runtime is polynomial.

The proof now invokes the approximation result appropriate for the paper’s supermodular contribution. Lemma 10 used Wolsey et al. (Wolsey, 1982) for monotone submodular functions; here we use Shi et al. (Shi et al., 2021), which gives an $\mathcal{O}\left(\frac{1}{1-k^f} + \ln\left(\frac{v_{\text{suff}}^g([n])}{\min_{i \in [n]} v_{\text{suff}}^g(\{i\})}\right)\right)$ guarantee for the supermodular setting, where $k^f := 1 - \min_{i \in [n]} \frac{v([n]) - v([n] \setminus \{i\})}{v(i) - v(\emptyset)}$. Proposition 2 gives monotonicity of v_{suff}^g , and Proposition 3 gives supermodularity under feature independence. Since Algorithm 2 runs in polynomial time, these properties yield the stated approximation factor, with $k^f := 1 - \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)}$.

To show that this expression is both bounded and constant, we follow the same preprocessing step as in Lemma 10, where we remove all redundant features from f — a process that, as previously explained, can be carried out in polynomial time for both neural networks and tree ensembles. As before, a feature i is strictly redundant if and only if $v(\{i\}) - v(\emptyset) = 0$. This preprocessing yields a new function f' that behaves identically to f over the remaining features and ensures that, for every i , both $v_{\text{suff}}^g(\{i\}) > 0$ and $v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset) > 0$ hold.

Now, following the same reasoning as in Lemma 10, where the probability term is lower bounded by $\frac{1}{|\mathcal{D}|}$ and its expected value by $\frac{1}{|\mathcal{D}|^2}$, we obtain that $\min_{i \in [n]} v_{\text{suff}}^g(i) \geq \frac{1}{|\mathcal{D}|^2}$ and $v_{\text{suff}}^g(i) - v_{\text{suff}}^g(\emptyset) \geq \frac{1}{|\mathcal{D}|^2}$. This implies, as in Lemma 10, that the first term is lower bounded by $\mathcal{O}(\ln(|\mathcal{D}|))$, and for the second term we derive the following:

Since v_{suff}^g is supermodular (Proposition 3), we have that $v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})$. It follows that $1 \leq v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\}) \leq \frac{1}{|\mathcal{D}|^2}$. This implies: $1 \leq \min_{i \in [n]} \frac{v_{\text{suff}}^g([n]) - v_{\text{suff}}^g([n] \setminus \{i\})}{v_{\text{suff}}^g(\{i\}) - v_{\text{suff}}^g(\emptyset)} \leq |\mathcal{D}|^2$, and therefore, $1 - |\mathcal{D}|^2 \leq k^f \leq 0$. This yields $\frac{1}{1-k^f} \leq |\mathcal{D}|^2$, demonstrating that the overall approximation bound based on the one established by Shi et al. is constant. While this bound may be significantly smaller in practice, our goal here is simply to show that it remains constant — unlike in the local setting, where no bounded approximation is achievable.

□

Lemma 13. *Unless $\text{PTIME} = \text{NP}$, there is no bounded approximation for computing a local cardinally minimal δ -sufficient reason for any $\langle f, \mathbf{x} \rangle$, even when $|\mathcal{D}| = 1$.*

Proof. The proof follows a similar approach to Lemma 11. Suppose, for contradiction, that there exists a polynomial-time algorithm that provides a bounded approximation to a cardinally minimal local δ -sufficient reason for some $\langle f, \mathbf{x} \rangle$, where f is either a neural network or a tree ensemble—even when $|\mathcal{D}| = 1$. Yet, Lemma 6 shows that even in the extreme case where the dataset consists of a single baseline $\mathbf{z} = \mathbf{0}_n$, deciding whether a sufficient reason exists is NP-hard unless $\text{PTIME} = \text{NP}$. Hence, the existence of such an approximation algorithm would contradict this hardness result, as it would entail the ability to decide the existence of a sufficient reason.

□

M HARDNESS OF APPROXIMATING GLOBAL CONTRASTIVE REASONS FOR NEURAL NETWORKS USING EMPIRICAL DISTRIBUTIONS

Proposition 1. *Assume a neural network $f : \mathbb{F} \rightarrow [c]$ with ReLU activations, some dataset $\mathcal{D}$, such that $\mathcal{D}$ is the empirical distribution defined over $\mathcal{D}$, and some threshold $\delta \in \mathbb{Q}$. Then unless $\text{NP} \subseteq \text{DTIME}(n^{\mathcal{O}(\log(\log(n)))})$, there does not exist a polynomial-time $\mathcal{O}(\ln(|\mathcal{D}|) - \epsilon)$ approximation algorithm for computing a cardinally minimal global contrastive reason for f concerning $\mathcal{D}$, for any fixed $\epsilon > 0$. This proposition shows that the logarithmic global approximation factor is essentially tight in the empirical-distribution setting.*

Proof. We prove tightness through a polynomial-time approximation-preserving reduction from a Set Cover variant that, like ordinary Set Cover, is hard to approximate within the factor stated in the proposition (Vazirani, 2001).

Partial Set Cover:

Input: A universe $U := \{u_1, \dots, u_n\}$, a collection of sets $S := \{S_1, \dots, S_m\}$ such that for all j $S_j \subseteq U$, and a coverage threshold $\delta \in \{0, 1, \dots, n\}$.

Output: A minimum-size subset $C \subseteq S$ such that $|\cup_{S_j \in C} S_j| \geq \delta$.

We prove the lower bound by reducing this problem to computing a *global* cardinally minimal contrastive reason S for a neural network f , with respect to an empirical distribution $\mathcal{D}$ over a dataset $\mathbf{D}$ and a threshold δ' . We begin by recalling the definition of a global contrastive reason and then extend it to the setting where the explanation is evaluated over an empirical dataset $\mathbf{D}$. The definition is as follows:

$$\mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] \geq \delta. \quad (73)$$

More specifically, when we assume an empirical distribution over $\mathbf{D}$, this implies that:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} [\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x})} \right) \end{aligned} \quad (74)$$

where we denote $\mathbf{D}_{\bar{S}}(\mathbf{z}, \mathbf{x}) := \{\mathbf{z} \in \mathbf{D} : \mathbf{z}_{\bar{S}} = \mathbf{x}_{\bar{S}}\}$. We now construct a ReLU neural network f . Before doing so, we specify the functional form that we intend the constructed neural network f to implement. For each set S_j , we introduce two vectors $\mathbf{x}^j$ and $\mathbf{y}^j$, both belonging to $\mathbb{Z}^n$ and drawn from the dataset $\mathbf{D}$, such that they agree on all coordinates outside S_j ; that is, they share identical feature values on the complement $\bar{S}_j$. Formally:

$$\mathbf{x}_{\bar{S}_j}^j = \mathbf{y}_{\bar{S}_j}^j \quad (75)$$

and we assign strictly different values to the features in S_j by selecting them from two distinct vectors $\mathbf{z}$ and $\mathbf{z}'$ such that:

$$\mathbf{x}^j = (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq (\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = (\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = \mathbf{y}^j \quad (76)$$

We aim to construct f such that it satisfies the following:

$$f(\mathbf{x}^j) = f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}^j) \neq f(\mathbf{x}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = f(\mathbf{y}_{\bar{S}_j}^j; \mathbf{z}_{S_j}'^j) = f(\mathbf{y}^j) \quad (77)$$

Let us now assume that both $\mathbf{x}^j$ and $\mathbf{y}^j$ occur *uniquely* in the dataset $\mathbf{D}$ across all $[n]$ coordinates — that is, no other vector in $\mathbf{D}$ matches either of them on any coordinate. Under this assumption, we obtain the following relation:

$$\begin{aligned} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}}[\Pr_{\mathbf{z} \sim \mathcal{D}}(f(\mathbf{z}) \neq f(\mathbf{x}) \mid \mathbf{z}_{\bar{S}_j} = \mathbf{x}_{\bar{S}_j})] &= \\ \frac{1}{|\mathbf{D}|} \sum_{\mathbf{x} \in \mathbf{D}} \left(\frac{\sum_{\mathbf{z} \in \mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \mathbf{1}_{\{f(\mathbf{z}) \neq f(\mathbf{x})\}}}{\mathbf{D}_{\bar{S}_j}(\mathbf{z}, \mathbf{x})} \right) &= \frac{2}{2|\mathbf{D}|} = \frac{1}{|\mathbf{D}|} \end{aligned} \quad (78)$$

Here, the factor of 2 appearing in both the numerator and denominator arises from the fact that only two vector pairs — $(\mathbf{x}^j, \mathbf{z}^j)$ and $(\mathbf{z}^j, \mathbf{x}^j)$ — contribute to the respective sums.

We repeat this construction for each S_i in the set-cover instance, each time choosing two *distinct* vectors $\mathbf{x}^i$ and $\mathbf{y}^i$ that differ on *every* coordinate in $[n]$. In total, this yields $2m$ vectors that form the dataset $\mathbf{D}$ from which we construct f .

The construction is implemented by a ReLU neural network f using a standard *neural network memorization* result (Vardi et al., 2022): a ReLU network representing any finite set of input–output pairs can be built in polynomial time. We use this representation for the constructed dataset and set $\delta' := \frac{\delta}{|\mathbf{D}|}$.

Finally, we connect the construction back to the approximation lower bound. A subset S_j is a global contrastive reason for f at tolerance δ' exactly when C is a partial set cover for U at tolerance δ . By Equation 78, each constructed vector pair $(\mathbf{x}^j, \mathbf{y}^j)$ contributes exactly $\frac{1}{|\mathbf{D}|}$ to the value term, and distinct S_j use disjoint vector pairs in $\mathbb{Z}^n$. Thus, set-cover coverage corresponds exactly to accumulated global contrastive value, completing the reduction.

□

N TRACTABILITY RESULTS FOR ORTHOGONAL DNFs

Lemma 14. *If f is an orthogonal DNFs and the probability term v_{suff}^g can be computed in polynomial time given the distribution $\mathcal{D}$ (which holds for independent distributions, among others), then obtaining a subset-minimal global δ -sufficient reason can be obtained in polynomial time.*

Proof. We first recall orthogonal DNFs (Crama & Hammer, 2011), since this structure is what lets the global value term be computed efficiently.

Definition 2 (Orthogonal DNFs). *A Boolean function φ is in orthogonal disjunctive normal form (orthogonal DNF) (Crama & Hammer, 2011) if it is a disjunction of terms $T_1 \vee T_2 \vee \dots \vee T_m$, such that for every pair of distinct terms T_i and T_j ($i \neq j$):*

$$T_i \wedge T_j \models \perp$$

This means that no single variable assignment can satisfy more than one term simultaneously.

The proof follows the same contribution pattern as Theorem 1: once v_{suff}^g is computable in polynomial time for any set S , the monotone global search yields a subset-minimal explanation efficiently.

For a given set S , let $\mathcal{C}^S T_i$ be the constraints that the term T_i imposes on the variables in S . We enumerate all pairs of terms (T_i, T_j) . Any pair whose constraints on S are *inconsistent* (i.e., $\mathcal{C}_{T_i}^S \wedge \mathcal{C}_{T_j}^S \models \perp$) is ignored, since it contributes zero to v_{suff}^g .

For each *consistent* pair of terms (T_i, T_j) , we evaluate $\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S)$, where $\mathcal{C}_{T_i, T_j}^S = \mathcal{C}_{T_i}^S \cap \mathcal{C}_{T_j}^S$. The contribution of this pair to v_{suff}^g is then

$$\Pr(\mathbf{x}_S \in \mathcal{C}_{T_i, T_j}^S) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_i}^{\bar{S}}) \cdot \Pr(\mathbf{x}_{\bar{S}} \in \mathcal{C}_{T_j}^{\bar{S}}).$$

Under feature independence, these probabilities factor as $\Pr(\mathbf{x}_S) = \prod_{i \in S} p(x_i)$. Because there are at most m^2 term pairs, this gives a polynomial-time computation of v_{suff}^g under an independent distribution $\mathcal{D}$, establishing the tractability extension. □

O DISCLOSURE: USAGE OF LLMs

An LLM was used only as a writing assistant for grammar, typo correction, and clarity. The research ideas, study design, data analysis, interpretation of results, and contribution claims were carried out solely by the authors.